

The Compass



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(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Subtraction without ‘Carry Over’

Material contributed by Krishaang Mishra

In performing subtraction of numbers do we always need to use the ‘carry’ procedure (also known as ‘exchanging’)? In this note, Krishaang shows that there is another way of doing subtraction, using negative numbers. He found the procedure on his own. We illustrate it using a few examples.

Example 1

Let’s see how to do $36 - 29$. We write the numbers in the usual manner:

$$\begin{array}{r} 36 \\ - 29 \\ \hline 1 \quad (-3) \end{array}$$

We see that 1 is in the tens’ place, and -3 in the units place. So the answer is $10 + (-3)$, which is the same as $10 - 3 = 7$. So the answer is 7.

Example 2

Let's see how to do $86 - 29$. We write the numbers as follows:

$$\begin{array}{r} 8 \quad 6 \\ - 2 \quad 9 \\ \hline 6 \quad (-3) \end{array}$$

In this case 6 is in the tens' place, and -3 in the units place. So the answer is $60 + (-3)$, which is the same as $60 - 3 = 57$. So the answer is 57.

Example 3

Let's see how to do $4258 - 3589$. We write the numbers as follows:

$$\begin{array}{r} 4 \quad 2 \quad 5 \quad 8 \\ - 3 \quad 5 \quad 8 \quad 9 \\ \hline 1 \quad (-3) \quad (-3) \quad (-1) \end{array}$$

To uncover the answer we do:

$$1000 - 300 = 700,$$

$$700 - 30 = 670,$$

$$670 - 1 = 669,$$

or in one line,

$$1000 - 3000 - 30 - 1 = 669.$$

So the answer is 669.

Example 4

Let's see how this works if there is a decimal point involved. Let's do $36.6 - 29.9$.

One approach is leave out the decimal point, carry out the computation as explained above, and then insert the decimal point at the end, in the correct place. Thus:

$$\begin{array}{r} 366 \\ - 299 \\ \hline 1 \quad (-3) \quad (-3) \end{array}$$

This yields:

$$100 - 30 - 3 = 100 - 33 = 67.$$

So the answer is 6.7.

Or we work with the decimal point:

$$\begin{array}{r} 36.6 \\ - 29.9 \\ \hline 1 \quad (-3) \quad (-.3) \end{array}$$

This yields:

$$10 - 3 - 0.3 = 6.7,$$

same as earlier.

Example 5

Now let's do $747.65 - 578.97$.

We have:

$$\begin{array}{r}
 7 \quad 4 \quad 7 \quad .6 \quad 5 \\
 - 5 \quad 7 \quad 8 \quad .9 \quad 7 \\
 \hline
 2 \quad (-3) \quad (-1) \quad (-.3) \quad (-.02)
 \end{array}$$

So the answer is:

$$200 - 30 - 1 - 0.3 - 0.02 = 169 - 0.32 = 168.68.$$

Remark

The general procedure can easily be worked out through these examples.

About the author

Krishang Mishra is at present a student of Class 6 in Amity International School Saket, New Delhi. He enjoys reading books, playing games and sports. He is a Brown Belt in Karate and aspires to be in the armed forces. He may be contacted at pooja.magoo@gmail.com.

Bhavik's equation

Article contributed by Bhavik Agrawal & Utkarsh Kumar

It happened once that while checking a student's notebook at school, I (Utkarsh Kumar) wondered if there were any natural number solutions to either of the following equations involving three variables:

$$a + \frac{b}{c} = a \times \frac{b}{c}, \quad (1)$$

and

$$a - \frac{b}{c} = a \times \frac{b}{c}. \quad (2)$$

In this short article we look at the first of these equations.

My student Bhavik obtained a solution by "hit-and-trial." Following this, he was able to provide a logical explanation for the particular case when $a = b$. He reasoned as follows.

In (11), suppose that $a = b$. We then get:

$$b + \frac{b}{c} = b \times \frac{b}{c},$$

$$\therefore \frac{bc + b}{c} = \frac{b^2}{c},$$

$$\therefore bc + b = b^2,$$

$$\therefore b(c + 1) = b^2,$$

$$\therefore c + 1 = b,$$

giving $c = b - 1$.

So, we have found a particular solution to (11):

Pick any value greater than 1 for b . Let $a = b$ and $c = b - 1$.

Then (a, b, c) satisfy (11).

E.g., let $b = 3$. Then $c = 3 - 1 = 2$. So $a = 3$, $b = 3$, $c = 2$. Let us check whether this works:

$$a + \frac{b}{c} = 3 + \frac{3}{2} = \frac{6+3}{2} = \frac{9}{2},$$

$$a \times \frac{b}{c} = 3 \times \frac{3}{2} = \frac{9}{2}.$$

We do have equality. The class was amazed with this finding!

As Bhavik's teacher, I was naturally interested in finding a general solution to the problem, i.e., a general solution in natural numbers without any condition of the kind $a = b$. In particular, I was interested in finding solutions where a, b, c are unequal.

After spending some time with the problem, I developed the following solution.

General solution of (11)

We have,

$$a + \frac{b}{c} = a \times \frac{b}{c},$$

$$\therefore \frac{ac + b}{c} = \frac{ab}{c},$$

$$\therefore ab - ac = b,$$

$$\therefore a(b - c) = b,$$

$$\therefore a = \frac{b}{b - c}.$$

Now for some reflection. Since a, b, c are natural numbers, $b - c$ must be a divisor of b . The simplest way of ensuring this is to put $b - c = 1$, i.e., $b = c + 1$. This yields:

$$a = \frac{b}{b - c} = \frac{c + 1}{1} = c + 1,$$

so the solution is:

Pick any natural number c . Let $a = c + 1$, $b = c + 1$. Then (a, b, c) satisfy equation (11).

Note that in this solution we get $a = b$. So, the question remains whether there are any solutions with $a \neq b$.

Yes, this is possible! For example, we may choose $b = 8$, $c = 6$, in which case we get

$$a = \frac{b}{b - c} = \frac{8}{8 - 6} = \frac{8}{2} = 4.$$

So we have $a = 4$, $b = 8$, $c = 6$, and

$$4 + \frac{8}{6} = 4 \times \frac{8}{6},$$

both sides being equal to $32/6$, i.e., $16/3$.

We invite you to find the general solution to the case $a \neq b$. You will find that if you impose the additional condition that a and b are coprime (i.e., they share no common factor other than 1), then you get the solution found above, with $b = c + 1$ and $a = b$. So, with this additional constraint, there is no solution with $a \neq b$.

The solution to (12) is very similar, and we invite you to find it.

About the authors

Bhavik Agrawal is a student of class 7. He is fond of math, especially problem-solving, and figuring out how it is used in everyday life. He also enjoys solving puzzles and playing chess in his spare time. He writes, “when learning is combined with creativity and curiosity, it becomes more exciting.”

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2135

Article contributed by Prithwiji De

Once while taking a stroll in the beautiful campus of Homi Bhabha Centre for Science Education (Mumbai), I came across a vehicle with a registration number 2135. Usually I look for prime factors of a number when I see one with not more than four digits. In this case it was rather simple because 21 and 35 are both multiples of 7 and so 7 must divide 2135, and the rest is easy ($2135 = 5 \times 7 \times 61$). But then I noticed something more.

Look at this: $2 + 3 = 1 \times 5$ and $2 \times 3 = 1 + 5$. This is nice. Can I find all 4-digit numbers with such a property?

Of course with the digits 1, 2, 3, 5, I get the eight numbers 1253, 1352, 2135, 2531, 3125, 3521, 5213, 5312 with the same property. And if I allow repetition of digits, then I immediately see that 0000 and 2222 are staring at me. But have I got them all? It is time for 'back of the envelope' calculations.

If the number I am looking for is $\overline{abcd}$ where a, b, c, d are digits, then the following equations must be satisfied: $a + c = bd$, $b + d = ac$.

Combining the two gives:

$$ac - (a + c) + bd - (b + d) = 0,$$

$$\therefore (a - 1)(c - 1) + (b - 1)(d - 1) = 2.$$

If the minimum of the four numbers a, b, c, d is 2, then so is the maximum. For if the maximum is at least 3, then

$$(a - 1)(c - 1) + (b - 1)(d - 1) > 2,$$

which is a contradiction. Thus $a = b = c = d = 2$, and the desired number is 2222.

If the minimum is 1, then either

$$(a - 1)(c - 1) = 0 \quad \text{or} \quad (b - 1)(d - 1) = 0.$$

In the former case $\{a, c\} = \{1, 5\}$, $\{b, d\} = \{2, 3\}$, and in the latter case, $\{a, c\} = \{2, 3\}$, $\{b, d\} = \{1, 5\}$. So there are no more such numbers than what I have listed above.

If the minimum is 0 then $a = b = c = d = 0$, which yields $\overline{abcd} = 0000$.

About the author

Prithwijit De is the National Coordinator of the Mathematical Olympiad Programme of the Government of India. He is an Associate Professor at the Homi Bhabha Centre for Science Education (HBCSE), TIFR, Mumbai. He loves to read and write popular articles in mathematics as much as he enjoys mathematical problem solving.

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Mirror Multi Pairs

Article contributed by Hara Gopal R

A pair of positive integers are said to be a *Mirror Multi Pair* if their product is unchanged if we reverse the digits. That is, suppose $\overline{ab}$ and $\overline{cd}$ are two 2-digit numbers such that

$$\overline{ab} \times \overline{cd} = \overline{ba} \times \overline{dc}.$$

Then we say that $\overline{ab}$ and $\overline{cd}$ are a “Mirror Multi Pair.”

Example. Observe:

$$23 \times 64 = 1472 = 32 \times 46.$$

As the product is unchanged if we reverse the digits, $\{23, 64\}$ is a “Mirror Multi Pair.”

Note The two numbers in the “Mirror Multi Pair” need not have the same number of digits. Here is an example involving a two-digit number and a

three-digit number:

$$24 \times 231 = 5544 = 42 \times 132.$$

Searching for mirror multi pairs

Let us see how find such pairs systematically.

The case of two-digit number pairs: Let $\overline{ab} = 10a + b$ and $\overline{cd} = 10c + d$ be such that

$$(10a + b) \times (10c + d) = (10b + a) \times (10d + c). \quad (3)$$

Expanding both sides we obtain

$$100ac + 10ad + 10bc + bd = 100bd + 10bc + 10ad + ac,$$

$$\therefore 99ac = 99bd,$$

$$\therefore ac = bd,$$

hence,

$$\frac{a}{d} = \frac{b}{c}. \quad (4)$$

So, if we take 4 digits a, b, c, d satisfying (12), then the numbers $\overline{ab}$ and $\overline{cd}$ formed by those digits are a Mirror Multi Pair.

We may now generate the 4-tuples (a, b, c, d) satisfying (12) and thus obtain Table 1. The Mirror Multi Pairs are (12,42); (12,63); (13,62); (12,84); (14,82); (13,93); (23,64); (24,63); (23,96); and (24,84).

a	b	c	d	$\overline{ab}$	$\overline{cd}$	$\overline{ba}$	$\overline{dc}$	Product
1	2	4	2	12	42	21	24	504
1	2	6	3	12	63	21	36	756
1	3	6	2	13	62	31	26	806
1	2	8	4	12	84	21	48	1008
1	4	8	2	14	82	41	28	1148
1	3	9	3	13	93	31	39	1209
2	3	6	4	23	64	32	46	1472
2	4	6	3	24	63	42	36	1512
2	3	9	6	23	96	32	69	2208
2	4	8	4	24	84	42	48	2016

Table 1: List of all two-digit mirror multi pairs

The case of two-digit number and a three-digit number: Let $10a + b$ and $100c + 10d + e$ be the two numbers. The condition then is

$$(10a + b) \times (100c + 10d + e) = (10b + a) \times (100e + 10d + c). \tag{5}$$

This leads to

$$\begin{aligned} &1000ac + 100ad + 10ae + 100bc + 10bd + be \\ &= 1000be + 100bd + 10bc + 100ae + 10ad + ac. \end{aligned}$$

Simplifying and dividing by 9, we get:

$$111(ac - be) + 10(ad - ae + bc - bd) = 0,$$

$$\therefore 111(ac - be) = 10(ae - ad + bd - bc)$$

Adding $10(ac - ae)$ on both sides, we get

$$121(ac - be) = 10(ae - ad + bd - bc + ac - be),$$

$$\therefore 121(ac - be) = 10(ae - be + bd - ad + ac - bc)$$

$$= 10 \left(e(a - b) - d(a - b) + c(a - b) \right),$$

$$\therefore 121(ac - be) = 10(a - b)(e - d + c). \quad (6)$$

We have not attempted to find the general solution of (14). Rather, we will only look for solutions in which both sides of (14) are 0. That is,

$$121(ac - be) = 0 \quad \text{and} \quad (a - b)(e - d + c) = 0.$$

As we do not want $a - b = 0$ (if $a = b$, then we will not get a different number when we reverse the digits), these lead to

$$ac - be = 0 \quad \text{and} \quad e - d + c = 0, \quad (7)$$

i.e., $a/e = b/c$ and $d = c + e$.

So, if we take 5 digits a, b, c, d, e such that they satisfy (15), then the numbers formed by those digits are Mirror Multi Pairs. In Table 2 we display some of these solutions.

a	b	c	d	e	$\overline{ab}$	$\overline{cde}$	$\overline{ba}$	$\overline{edc}$	Product
1	2	4	6	2	12	462	21	264	5544
1	2	6	9	3	12	693	21	396	8316
1	3	6	8	2	13	682	31	286	8866
2	4	2	3	1	24	231	42	132	5544
2	4	6	9	3	24	693	42	396	16632
3	6	2	3	1	36	231	63	132	8316

Table 2

From Table 2 we see that the two- and three-digit Mirror Multi Pairs are (12,462); (12,693); (13,682); (24,231); (24,693); and (36,231)..

Interested readers could try to find the general solution of (14).

The case of a pair of three-digit numbers: Let $100a + 10b + c$ and $100d + 10e + f$ be the two numbers. The condition then is

$$\begin{aligned} &(100a + 10b + c) \times (100d + 10e + f) \\ &= (100c + 10b + a) \times (100f + 10e + d). \end{aligned}$$

On simplification we get the following relation:

$$a(101d + 10e) + 10b(d - f) - c(10e + 101f) = 0. \tag{8}$$

Rewriting this as

$$101(ad - cf) + 10(ae + bd - bf - ce) = 0, \tag{9}$$

we see that a possible solution is obtained by imposing the following additional condition:

$$\frac{a}{f} = \frac{b}{e} = \frac{c}{d}. \tag{10}$$

For, by writing $a = kf, b = ke, c = de$ for some number k and substituting in (64), we get

$$101k(df - df) + 10k(ef + de - ef - de),$$

which simplifies to 0, as required. So, if we take six digits a, b, c, d, e, f satisfying (65), then the numbers formed by those digits are a three-digit Mirror Multi Pair. Table 3 displays solutions obtained this way.

a	b	c	d	e	f	$\overline{abc}$	$\overline{def}$	$\overline{cba}$	$\overline{fed}$	Product
1	2	3	9	6	3	123	963	321	369	118449
1	2	3	6	4	2	123	642	321	246	78966
1	2	4	8	4	2	124	842	421	248	104408
2	3	4	8	6	4	234	864	432	468	202176
2	4	1	2	8	4	241	284	142	482	68444
2	4	6	9	6	3	246	963	642	369	236898

Table 3: Some three-digit mirror multi pairs

So, (123,963); (123,642); (124,842); (234,864); (241,284); and (246,963) are some three-digit mirror multi pairs.

As earlier, we have not tried to find the general solution of (64). Readers may want to give it a try.

Conclusion

We can continue this exploration to any extent, to discover various types of Mirror Multi Pairs of numbers.

About the author

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A pretty property of the pair of numbers 34 and 67

Playing around with the numbers 34 and 67, we discover this relation:

$$34^3 + 67^3 = 340067.$$

Please verify this. Then we discover something more striking:

$$334^3 + 667^3 = 334000667.$$

We quickly sense that this is the beginning of a pattern that persists:

$$34^3 + 67^3 = 340067,$$

$$334^3 + 667^3 = 334000667,$$

$$3334^3 + 6667^3 = 333400006667,$$

$$33334^3 + 66667^3 = 333340000066667,$$

and so on.

But can we be sure of “and so on”? Does the pattern really persist forever? We cannot be sure unless we prove it rigorously. Here is a proof based on the elegant demonstration in [1].

For the equality $34^3 + 67^3 = 340067$, write $a = 34$ and $b = 67$. Observe that

$$a = \frac{10^2 - 1}{3} + 1 = \frac{10^2 + 2}{3},$$

$$\therefore 10^2 = 3a - 2.$$

$$\text{Also, } b = 2a - 1.$$

Hence we have:

$$\begin{aligned} 34^3 + 67^3 &= a^3 + b^3 = a^3 + (2a - 1)^3 \\ &= 9a^3 - 12a^2 + 6a - 1, \end{aligned}$$

$$\begin{aligned} \text{and also, } 340067 &= 10^4 a + b = (3a - 2)^2 a + (2a - 1) \\ &= 9a^3 - 12a^2 + 6a - 1. \end{aligned}$$

From the equality of the expressions thus obtained, we understand the algebra behind the equality $34^3 + 67^3 = 340067$.

Let's consider next the equality $334^3 + 667^3 = 334000667$. As earlier let us write $a = 334$ and $b = 667$. Then we have:

$$a = \frac{10^3 - 1}{3} + 1 = \frac{10^3 + 2}{3},$$

$$\therefore 10^3 = 3a - 2.$$

$$\text{Also, } b = 2a - 1.$$

Hence we have:

$$\begin{aligned} 334^3 + 667^3 &= a^3 + b^3 = a^3 + (2a - 1)^3 \\ &= 9a^3 - 12a^2 + 6a - 1, \end{aligned}$$

$$\begin{aligned} \text{and also, } 334000667 &= 10^6a + b = (3a - 2)^2a + (2a - 1) \\ &= 9a^3 - 12a^2 + 6a - 1. \end{aligned}$$

We now understand the algebra behind the equality $334^3 + 667^3 = 334000667$.

Observe that the expressions obtained at the end are identical to those we got earlier! This tells us how to proceed in the general case.

Let $a = 333\dots34$, the number of 3's being $(n - 1)$, and $b = 666\dots67$, the number of 6's being $(n - 1)$. Then,

$$a = 333\dots34 = \frac{10^n - 1}{3} + 1 = \frac{10^n + 2}{3},$$

$$\therefore 10^n = 3a - 2.$$

$$\text{Also, } b = 666\dots67 = 2a - 1.$$

Hence,

$$a^3 + b^3 = a^3 + (2a - 1)^3 = 9a^3 - 12a^2 + 6a - 1.$$

Next, consider the number $333\dots34000\dots00666\dots67$, the number of 0's being n .

After a careful examination of the place-value structure of the number, we proceed as follows.

We have,

$$\begin{aligned}
 333\dots34000\dots00666\dots67 &= 333\dots34 \times 10^{2n} + 666\dots67 \\
 &= a \times 10^{2n} + (2a - 1) \\
 &= a \times (10^n)^2 + (2a - 1) \\
 &= a(3a - 2)^2 + (2a - 1) \\
 &= 9a^3 - 12a^2 + 6a - 1.
 \end{aligned}$$

We have obtained the same expression as earlier. Hence the stated result is true. ■

Remark: A computer assisted search reveals that $\{34, 67\}$ is the only pair of two-digit numbers with this property, and $\{334, 667\}$ is the only such three-digit pair.

Further reading

- [1] Alon Amit, “ $34^3 + 67^3 = 340067$. $334^3 + 667^3 = 334000667$. Does this relation continue forever?” from <https://www.quora.com/34-3-67-3-340067-334-3-667-3-334000667-3334-3-6667-3-333400006667-Does-this-relation-continue-forever/answer/Alon-Amit>

Article contributed by Shailesh Shirali

A Reverse-and-Subtract Iteration

Here is an interesting digit-based iteration that children may chance upon if they are in the habit of “mathematical doodling.”

Notation: For any integer n , let $f(n)$ denote the integer obtained by writing the digits of n in reverse order. For example, $f(35) = 53$, $f(103) = 301$, $f(20) = 2$.

Take any positive integer n . Replace it by $|n - f(n)|$. If the answer is 0, then terminate the iteration. Else, repeat the same operation with the new number.

Experimental finding: We always seem to reach 0 in a few steps.

Is this really so? Will we always reach 0, no matter where we start?

Let's explore ... First, some examples.

Examples

- If we start with 12, we reach 0 in 2 steps:

$$12 \mapsto 9 \mapsto 0.$$

- If we start with 16, we reach 0 in 3 steps:

$$16 \mapsto 45 \mapsto 9 \mapsto 0.$$

- If we start with 14, we reach 0 in 4 steps:

$$14 \mapsto 27 \mapsto 45 \mapsto 9 \mapsto 0.$$

- If we start with 15, we reach 0 in 5 steps:

$$15 \mapsto 36 \mapsto 27 \mapsto 45 \mapsto 9 \mapsto 0.$$

- If we start with 62, we reach 0 in 5 steps:

$$62 \mapsto 36 \mapsto 27 \mapsto 45 \mapsto 9 \mapsto 0.$$

- If we start with 13, we reach 0 in 6 steps:

$$13 \mapsto 18 \mapsto 63 \mapsto 27 \mapsto 45 \mapsto 9 \mapsto 0.$$

- If we start with 113, we reach 0 in 6 steps:

$$113 \mapsto 198 \mapsto 693 \mapsto 297 \mapsto 495 \mapsto 99 \mapsto 0.$$

If we continue this experimentation for all starting numbers up to 1000, we find that we reach 0 in every single case, never taking more than 6 steps!

At this point we are ready to conclude that we will reach 0 at some point, no matter what number we start with. We are now ready to conjecture that this will always be the case. (We might even conjecture that we will reach 0 within 6 steps, no matter what number we start with.)

And then, to our surprise, we discover that the conjecture is false!

First, we find that there are numbers that take more than 6 steps to reach 0. One such is the number 1011. Starting with this number, we obtain:

$$1011 \mapsto 90 \mapsto 81 \mapsto 63 \mapsto 27 \mapsto 45 \mapsto 9 \mapsto 0.$$

So, it takes 7 steps in this case. (The fact that this part of the conjecture has failed may not come as much of a surprise. It was surely too ambitious of us to imagine that we will always reach zero within 6 steps.)

The real surprise comes with the very next number: 1012. We find that the sequence gets caught in a cycle without ever reaching zero! Namely:

$$1012 \mapsto 1089 \mapsto 8712 \mapsto 6534 \mapsto 2178 \mapsto 6534 \mapsto 2178 \mapsto \dots$$

Observe that the sequence moves from 6534 to 2178 and then back to 6534. As it has reached the same number again, we can say with confidence that the sequence will never reach 0 at all. It has instead got caught in a 2-cycle.

Moral of this episode! We see from this simple experience how easy it is to be misled by data and to launch into false conjectures.

Article contributed by Shailesh Shirali

Playing with Primitive Pythagorean Triples – 5

We continue our exploration of Primitive Pythagorean Triples (PPTs). In this article, we shall explore the following two strands:

- (a) Looking for PPTs (a, b, c) in which a and b are consecutive numbers, i.e., PPTs of the form $(a, a + 1, c)$;
- (b) Looking for PPTs (a, b, c) in which some two of a, b, c are prime numbers, e.g., the PPT $(5, 12, 13)$.

PPTs of the form $(a, a+1, c)$

We have earlier studied PPTs of the form $(a, b, b + 1)$, i.e., PPTs (a, b, c) in which b and c are consecutive numbers. These turn out to be simple to generate and classify completely. Triples of the form $(a, a + 1), c$, i.e., PPTs (a, b, c) in which a and b are consecutive numbers, are trickier to find but hold a great deal of interesting mathematics within them. We introduce

the problem in this section but do not solve it fully as there is an article on this very theme immediately following this article.

Let a, c be positive integers such that

$$a^2 + (a + 1)^2 = c^2. \quad (11)$$

This yields $2a^2 + 2a + 1 = c^2$, leading to

$$\begin{aligned} 4a^2 + 4a + 2 &= 2c^2, \\ \therefore (2a + 1)^2 + 1 &= 2c^2, \\ \therefore (2a + 1)^2 - 2c^2 &= -1. \end{aligned} \quad (12)$$

The equation in the last line is of the form

$$x^2 - 2y^2 = \pm 1, \quad (13)$$

with $x = 2a + 1$, $y = c$, and with the minus sign. This equation is often referred to as *Pell's equation*, after the 17th century English mathematician John Pell (1611–1685)[1] who wrote about it. (The name came about because of Euler's attributing the analysis to him. But unknown to Euler and all other European mathematicians was the fact that "Pell's equation" had been explored at great depth many centuries earlier in India, by the mathematicians Brahmagupta (7th century CE) and Bhaskaracharya II (12th century CE).)

We now study equation (60) computationally. For another viewpoint and analysis, refer to the article by Skandan immediately following this one.

The most straightforward approach now is to generate positive integer pairs (x, y) that satisfy (60) and then to examine the pairs for patterns.

We have used a computer to generate the following table of x, y -values. In the last row we show the values of $z = x^2 - 2y^2$.

x	1	3	7	17	41	99	239	...
y	1	2	5	12	29	70	169	...
z	-1	1	-1	1	-1	1	-1	...

The patterns in this table are striking. In the z -row we have alternating ± 1 . And the x -row and y -row show an identical connection across columns. That is, if x_n and y_n denote the n^{th} entries in the x -row and y -row (so $x_1 = 1$, $y_1 = 1$, $x_2 = 3$, $y_2 = 2$, $x_3 = 7$, $y_3 = 5$, ...), then for $n > 1, 2, 3, \dots$,

$$\left. \begin{aligned} x_{n+1} &= x_n + 2y_n, \\ y_{n+1} &= x_n + y_n, \end{aligned} \right\} \quad (14)$$

and,

$$\left. \begin{aligned} x_{n+2} &= 2x_{n+1} + x_n, \\ y_{n+2} &= 2y_{n+1} + y_n. \end{aligned} \right\} \quad (15)$$

Please verify for yourself that the patterns hold for all the numbers shown in the table.

Two questions immediately come to mind:

- (i) If we continue computing the terms of the sequence according to (14) and (15), will all the pairs (x_n, y_n) satisfy the equation $x^2 - 2y^2 = \pm 1$?
- (ii) Will this procedure capture all possible positive integer solutions to equation (60)?

The brief answers to both questions are Yes and Yes. But we will show why only for question (i). (We leave question (ii) for future consideration.)

The obvious way to proceed is to use the principle of induction. We shall now show that if the pair (x_n, y_n) satisfies (60), then so do the pairs

$$(x_{n+1}, y_{n+1}) \quad \text{and} \quad (x_{n+2}, y_{n+2})$$

given by (14) and (15).

Let a, b, c, d be integers such that

$$a^2 - 2b^2 = \pm 1, \quad c = a + 2b, \quad d = a + b. \quad (16)$$

We shall show that

$$c^2 - 2d^2 = -(a^2 - 2b^2), \quad (17)$$

and,

$$(2c + a)^2 - 2(2d + b)^2 = a^2 - 2b^2. \quad (18)$$

The relations follow in a straightforward manner:

$$\begin{aligned} c^2 - 2d^2 &= (a + 2b)^2 - 2(a + b)^2 \\ &= (a^2 + 4ab + 4b^2) - 2(a^2 + 2ab + b^2) \\ &= 2b^2 - a^2 = -(a^2 - 2b^2), \end{aligned}$$

and

$$\begin{aligned} (2c + a)^2 - 2(2d + b)^2 &= (3a + 4b)^2 - 2(2a + 3b)^2 \\ &= (9a^2 + 24ab + 16b^2) - 2(4a^2 + 12ab + 9b^2) \\ &= a^2 - 2b^2. \end{aligned}$$

These implications show that if the pair (x_n, y_n) satisfies (60), then so do the pairs (x_{n+1}, y_{n+1}) and (x_{n+2}, y_{n+2}) given by (14) and (15). This answers question (i).

An immediate corollary of this is that equation (60) has infinitely many solutions.

It follows in turn that there are infinitely many PPTs of the form $(a, a+1, c)$. To display them we first extract from the table shown above only those columns for which $x^2 - 2y^2 = -1$. We obtain:

x	1	7	41	239	1393	...
y	1	5	29	169	985	...

Recalling that we had made the substitutions $x = 2a + 1$, $y = c$, we obtain the following PPTs (we discard the solution $x = 1$, $y = 1$ as it leads to the triple $(0, 1, 1)$ which is of no interest):

$$\begin{array}{ll}
 (3, 4, 5) & (\text{from } x = 7, y = 5), \\
 (20, 21, 29) & (\text{from } x = 41, y = 29), \\
 (119, 120, 169) & (\text{from } x = 239, y = 169), \\
 (696, 697, 985) & (\text{from } x = 1393, y = 985),
 \end{array}$$

and so on. We can continue this list for as long as we wish.

There is more to say about this problem and various number sequences that emerge from it. However, we shall not say more for now. For another treatment of this problem, please read the article by Skandan.

Question: Examining the (x, y) pairs displayed in the table, we find that they yield excellent rational approximations to $\sqrt{2}$. Namely,

$$\frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \frac{99}{70}, \dots$$

E.g., $99/70 \approx 1.4143$, $\sqrt{2} \approx 1.4142$. How do you explain this?

PPTs with two prime numbers

Can there be PPTs in which *all three* numbers are prime? We know from our earlier studies that in a PPT (a, b, c) , one of a, b is odd and the other one is even, and c is odd. The even number in $\{a, b\}$ would have to be 2, as that is the only prime which is even. But there is no PPT of the $(2, b, c)$. This is easy to prove. (Please complete the proof on your own.)

What about PPTs in which *just one* of the three numbers is prime? We have two possibilities: PPTs (p, b, c) where p is prime ($p > 2$) but b, c are composite; PPTs (a, b, p) where p is prime but a, b are composite. Let's look at these two possibilities separately.

The PPT (p, b, c) where p is an odd prime: In this case we have

$$p^2 + b^2 = c^2, \quad \therefore p^2 = (c - b)(c + b).$$

The only way of expressing p^2 as a product of two unequal numbers is $1 \times p^2$, hence we have

$$c - b = 1, \quad c + b = p^2,$$

leading to

$$b = \frac{p^2 - 1}{2}, \quad c = \frac{p^2 + 1}{2}.$$

For $p = 3$ we get the PPT $(3, 4, 5)$; but note that this has primes and not one.

If $p > 3$, then b cannot be prime, since

$$\frac{p^2 - 1}{2} = \frac{(p - 1)(p + 1)}{2},$$

and both $\frac{1}{2}(p - 1)$ and $\frac{1}{2}(p + 1)$ exceed 1. (Or note that $b > 3$ and b is a multiple of 3, hence composite.)

But nothing can be said about c ; it could be prime or composite. Here is a list of primes p , $3 < p < 100$ for which $\frac{1}{2}(p^2 + 1)$ too is prime:

$$5, 11, 19, 29, 59, 61, 71, 79.$$

Thus, $\frac{1}{2}(5^2 + 1) = 13$, $\frac{1}{2}(11^2 + 1) = 61$, $\frac{1}{2}(19^2 + 1) = 181$, etc. Are there infinitely many primes p for which $\frac{1}{2}(p^2 + 1)$ too is prime. *I believe this is an open problem. It is unknown whether there are infinitely many such primes.*

On the other hand, it is easy to show that there are infinitely many primes p for which $\frac{1}{2}(p^2 + 1)$ is **not** prime. We only need to select primes $p > 3$ whose units digit is 3 or 7. This ensures that the units digit of p^2 is 9, hence that $\frac{1}{2}(p^2 + 1)$ is a multiple of 5 and greater than 5, hence not prime. It is known that there are infinitely many primes $p > 3$ whose units digit is 3 or 7. (We must say more about the “it is known” claim in this sentence. We will do so in a subsequent article.) It follows that there are infinitely many PPTs in which exactly one of the three numbers is prime.

We have now shown that for any prime $p > 3$,

- (i) there are no PPTs (p, b, c) where both b and c are prime;
- (ii) there are infinitely many PPTs (p, b, c) where both b and c are composite.

What about the case when only c is prime? As pointed out above, we do not know whether there are infinitely many such PPTs. (It is believed that there are infinitely many such cases, but there is no proof of this at present.) So this part of the question remains open.

It remains to consider PPTs of the form (a, b, p) . Our task is to solve the equation

$$a^2 + b^2 = p^2 \tag{19}$$

when p is prime. Quite remarkably, the following is true.

- (i) If $p \equiv 3 \pmod{4}$, then (66) has **no** solutions in positive integers.
- (ii) If $p \equiv 1 \pmod{4}$, then (66) **always has** a solution. We display a few instances of this below.

p	5	13	17	29	37	41
(a, b)	(3, 4)	(5, 12)	(8, 15)	(20, 21)	(12, 35)	(9, 40)

Exercise for the reader

Find solutions (a, b) to $a^2 + b^2 = p^2$ for the following cases: $p = 53$; $p = 61$; $p = 73$; $p = 89$; $p = 97$.

Closing note

There are several matters left incomplete in this article. We need to say more about the following statements: (i) There are infinitely many primes $p > 3$ whose units digit is 3 or 7. (ii) If $p \equiv 3 \pmod{4}$, then there do not exist positive integers a, b such that $a^2 + b^2 = p^2$. (iii) If $p \equiv 1 \pmod{4}$, then there do exist positive integers a, b such that $a^2 + b^2 = p^2$. We shall say more about these claims in the January 2026 issue.

Further reading

- [1] MacTutor, "John Pell" from <https://mathshistory.st-andrews.ac.uk/Biographies/Pell/>

Primitive Pythagorean Triplets and Skandan Numbers

Article contributed by Skandan Suresh

Introduction

In this article we study Pythagorean triplets of the form $(x, x + 1, z)$. Here are the first few examples of these triplets:

$$(3, 4, 5), \quad (20, 21, 29), \quad (119, 120, 169), \quad (696, 697, 985).$$

We see that the numbers increase rapidly. Are there easy ways to generate such triples? Are there an infinite number of them? In this article we describe a way to generate these triples, and this proves that there are an infinite number of them.

Brahmagupta's formula for generating PPTs

This is a formula for generating all Primitive Pythagorean Triplets (PPTs for short). If x, y, z are coprime positive integers such that $x^2 + y^2 = z^2$, then there exist coprime positive integers m, n (one odd and the other even) such that

$$x = m^2 - n^2, \quad y = 2mn, \quad z = m^2 + n^2, \quad (20)$$

OR

$$x = 2mn, \quad y = m^2 - n^2, \quad z = m^2 + n^2. \quad (21)$$

All PPTs can be expressed in this form. The triplets we are considering are primitive as x and $x + 1$ are necessarily coprime. Therefore, this is an appropriate formula to generate all such triplets.

Introducing Skandan's number

Generating PPTs of the form $(x, x + 1, z)$ involves a special kind of number series. We shall refer to numbers in this series as *Skandan's numbers*. To see how they arise, we use the Brahmagupta formula.

If $x^2 + (x + 1)^2 = z^2$, then there exist coprime positive integers m, n (one odd and the other even) such that

$$x = m^2 - n^2, \quad x + 1 = 2mn, \quad z = m^2 + n^2, \quad (22)$$

OR

$$x = 2mn, \quad x + 1 = m^2 - n^2, \quad z = m^2 + n^2. \quad (23)$$

In the first case we obtain, $m^2 - n^2 - 2mn = -1$,

$$\therefore n^2 + 2mn = m^2 + 1,$$

$$\therefore m^2 + 2mn + n^2 = 2m^2 + 1,$$

$$\therefore (m + n)^2 = 2m^2 + 1. \quad (24)$$

In the second case we obtain, $m^2 - n^2 - 2mn = 1$,

$$\therefore m^2 - 2mn = n^2 + 1,$$

$$\therefore m^2 - 2mn + n^2 = 2n^2 + 1,$$

$$\therefore (m - n)^2 = 2n^2 + 1. \quad (25)$$

We see that in both cases we get an equation of the form

$$S^2 = 1 + 2q^2 \quad (26)$$

where S and q are positive integers ($S = m + n$ in the 1st case, and $S = m - n$ in the 2nd case).

Claim: If we get numbers S and q satisfying (26), we can generate two Pythagorean triplets of the form $(x, x + 1, z)$.

Let's confirm.

Let S, q satisfy (26). Clearly, $S > q$.

We can now compute m, n in two ways:

- $m = q = \sqrt{(S^2 - 1)/2}, \quad n = S - m;$
- $n = q = \sqrt{(S^2 - 1)/2}, \quad m = S + n.$

The two (m, n) pairs now yield two PPTs via the Brahmagupta's formula.

Example One solution to (26) is $S = 3, q = 2$. These yield two (m, n) pairs:

- $m = 2, n = 1$, giving $m^2 - n^2 = 3, 2mn = 4, m^2 + n^2 = 5$.

We obtain the PPT $(3, 4, 5)$.

- $n = 2, m = 5$, giving $m^2 - n^2 = 21, 2mn = 20, m^2 + n^2 = 29$.

We obtain the PPT $(20, 21, 29)$.

Both PPTs are of the required form.

We may check algebraically that the triplets will always have the needed property (i.e., that the first two numbers differ by 1):

- In the first case we get

$$\begin{aligned} 2mn - (m^2 - n^2) &= m^2 + 2mn + n^2 - 2m^2 \\ &= (m + n)^2 - 2m^2 \\ &= S^2 - 2q^2 = 1. \end{aligned}$$

- In the second case we get

$$\begin{aligned} (m^2 - n^2) - 2mn &= m^2 - 2mn + n^2 - 2n^2 \\ &= (m - n)^2 - 2n^2 \\ &= S^2 - 2q^2 = 1. \end{aligned}$$

Hence, by finding a pair (S, q) with property (26), we are able to generate two PPTs with the needed property.

It follows that if we can generate infinitely many such pairs (S, q) , then we can generate infinitely many PPTs of the type $(x, x + 1, z)$.

Definition: A *Skandan number* is a positive integer $S > 1$ with the property that $\frac{1}{2}(S^2 - 1)$ is a perfect square.

For example, 3 and 17 have this property (they are the two smallest positive integers with the property):

$$\frac{3^2 - 1}{2} = 4 = 2^2, \quad \frac{17^2 - 1}{2} = 144 = 12^2.$$

Generating Skandan Numbers

We now consider the problem of generating all Skandan numbers. Let S be a Skandan number, say

$$S^2 = 1 + 2q^2$$

for some positive integer q . Then $S^2 - 1 = 2q^2$, so,

$$(S - 1)(S + 1) = 2q^2. \quad (27)$$

Since $2q^2$ is even, $S^2 - 1$ is even, so S is odd, so both $S - 1$ and $S + 1$ are even. Indeed, they are consecutive even numbers; one of them is divisible by 4, while the other one is twice an odd number. This means that their GCD is 2; they do not share any prime factor other than 2.

There are thus two possible situations which yield a solution for (27):

- (a) The first possibility is that $S + 1$ is a multiple of 4, and $S - 1$ is twice an odd number. In this case we may write

$$(S + 1) \cdot \left(\frac{S - 1}{2} \right) = q^2. \quad (28)$$

Since $S + 1$ and $\frac{1}{2}(S - 1)$ are coprime, (28) implies that both $S + 1$ and $\frac{1}{2}(S - 1)$ are perfect squares, say

$$S + 1 = A^2, \quad \frac{S - 1}{2} = B^2, \quad (29)$$

where A, B positive integers.

- (b) The second possibility is that $S - 1$ is a multiple of 4, and $S + 1$ is twice an odd number. In this case we may write

$$(S - 1) \cdot \left(\frac{S + 1}{2} \right) = q^2. \quad (30)$$

Since $S - 1$ and $\frac{1}{2}(S + 1)$ are coprime, (28) implies that both $S - 1$ and $\frac{1}{2}(S + 1)$ are perfect squares, say

$$S - 1 = A^2, \quad \frac{S + 1}{2} = B^2, \quad (31)$$

where A, B positive integers.

Producing more Skandan numbers

From (31) there follows an interesting development.

Since $S - 1 = A^2$ and $S + 1 = 2B^2$, we must have

$$A^2 = 2B^2 - 2.$$

Hence A is even. Let $A = 2C$, where C is a positive integer. Then $A^2 = 4C^2$, giving $4C^2 = 2B^2 - 2$, hence $B^2 = 2C^2 + 1$.

But this means that B is a Skandan number.

From (31) we also get $S = 2B^2 - 1$. This tells us that it is possible to generate infinitely many Skandan numbers starting from any Skandan number, by iterating this formula repeatedly:

$$S_{\text{new}} = 2(S_{\text{old}})^2 - 1. \quad (32)$$

Thus we have:

$$\begin{aligned} 3 &\mapsto 2(3^2) - 1 = 17 \\ &\mapsto 2(17^2) - 1 = 577 \\ &\mapsto 2(577^2) - 1 = 665857 \\ &\mapsto 2(665857)^2 - 1 = 886731088897 \mapsto \dots \end{aligned}$$

We may verify that 17, 577, 665857, 886731088897 are all Skandan numbers. Obviously, we can continue this progression for ever. This proves that there are infinitely many PPTs of the type $(x, x + 1, z)$.

But note that the mechanism described above does *not* generate all possible Skandan numbers. For example, 99 is a Skandan number, but it is not captured by the above formula.

Examining the other possibility

Will the other possibility yield more Skandan numbers? From (29) we get, since $S + 1 = A^2$ and $S - 1 = 2B^2$,

$$A^2 = 2B^2 + 2. \quad (33)$$

Hence A is even. Let $A = 2C$, where C is a positive integer. If B were even, then the left side of (33) would be $0 \pmod{4}$, as A is even, while the right

side would be $2 \pmod{4}$; not possible. Hence B is odd. Let $B = 2D + 1$, where D is a positive integer.

Now let us write (33) as $4C^2 - 4 = 2(2D + 1)^2 - 2$. Simplifying, we get

$$(C - 1)(C + 1) = 2D(D + 1). \quad (34)$$

As $2D(D + 1)$ is even, C must be odd. Let $C = 2E + 1$. Substituting in (34) we get $2E(2E + 2) = 2D(D + 1)$, giving

$$D(D + 1) = 2E(E + 1). \quad (35)$$

Now consider the original context that gave rise to this exploration: PPTs of the form $(x, x + 1, z)$. From the defining condition we get

$$x^2 + (x + 1)^2 = z^2,$$

$$\therefore 2x^2 + 2x = z^2 - 1,$$

$$\therefore 2x(x + 1) = (z - 1)(z + 1).$$

We see that z is odd; let $z = 2u + 1$. This yields $2x(x + 1) = 4u(u + 1)$, or

$$x(x + 1) = 2u(u + 1). \quad (36)$$

Observe the close similarity between (35) and (36). It tells us that from a PPT $(x, x + 1, z)$, we should be able to produce a Skandan number, by computing the various quantities and tracing back! We only have to keep in mind the following connecting equalities:

$$B = 2D + 1 \quad \text{and} \quad A = 2(2E + 1), \quad (37)$$

and

$$z = 2u + 1. \quad (38)$$

Examples

Start with the PPT (3,4,5). Here $x = 3$ and $z = 5$, hence $u = 2$. So let's put $D = 3$ and $E = 2$ in (35). This yields, from (37),

$$B = 7, \quad A = 10,$$

and $S = 10^2 - 1 = 99$.

Similarly, from the PPT (20,21,29) we get, in turn, $x = 20$, $z = 29$, $u = 14$, $D = 20$, $E = 14$, $B = 41$, $A = 58$, and

$$S = 58^2 - 1 = 3363.$$

We may verify that this is a Skandan number:

$$\frac{3363^2 - 1}{2} = 5654884 = 2378^2.$$

In general, from the PPT $(x, x+1, z)$ we obtain the Skandan number

$$S = \frac{z^2 - 1}{2}. \quad (39)$$

Acknowledgement

This article was motivated by a problem posed in a maths class (*Ganit Manthan*). Our teacher had got this problem when another student (Vivaan Jeedigunta) from a different batch asked,

“We know a way to generate PPTs (a, b, c) where b and c are consecutive. Is there a way to generate PPTs where a and b are consecutive?”

About the author

Skandan Suresh studies in Grade 8 at Prakriti School Noida. He has a keen interest in mathematics and has been pursuing this by joining various maths camps. He also has an avid interest in music. He has been learning to play the flute in the Carnatic style and has performed at various opportunities including the G20 summit 2023. Skandan is a sports enthusiast too. He loves to read books and travel.

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From Desks to Walls: Building Thinking Classrooms

Article contributed by Shiv Gaur

“The room was buzzing. Teachers — both math and non-math — were on their feet, arguing, discussing, and scribbling on whiteboards. All of this over a math problem.”

That scene unfolded a couple of years ago when a renowned math consultant visited our school. She had asked us to solve math problems, in small teams, on small vertical whiteboards, and I was struck by the level of engagement from everyone — even those who didn’t normally teach math. The sheer level of engagement left me intrigued and eager to understand why it worked so well.

Before she left, I asked her about the idea behind these vertical whiteboards. She recommended that I read Peter Liljedahl’s *Building Thinking Classrooms*.

Building Thinking Classrooms (BTC) is not a fixed curriculum but a framework of research-based practices that shape how students engage with

mathematics. Instead of focusing on content delivery, it emphasizes how learning is structured. Some of the key principles of this approach are:

Tasks Begin lessons with rich, non-routine thinking tasks (also known as “low-floor, high-ceiling” or “low threshold” tasks).

Student Grouping Place students in random groups of two or three.

Work Surfaces Have students work at vertical non-permanent surfaces (VNPS), such as whiteboards or chalkboards.

Flow of the Lesson Students stand and work collaboratively on tasks while the teacher circulates, observes, and asks guiding questions rather than providing direct instruction.

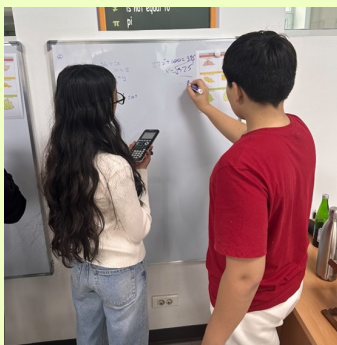
The only low-tech equipment needed to implement this approach is a set of small whiteboards with markers and erasers. I had eight boards installed along the side walls of my classroom, evenly spaced.

I began using them for small-group activities when introducing new concepts, for mathematical investigations, and even for some procedural practice. The grades involved were 8, 9, and 11. I noticed a significant increase in engagement across all grade levels. Because students were standing, moving, and working together, they were far more active than when seated at their desks. I also sensed that students were more willing to take risks: they could try an idea, erase it, and start again without hesitation. The temporary nature of the boards seemed to encourage experimentation. I also observed a lot of collaborative discussion within groups. As I circulated, I could easily see students’ mathematical thinking, identify misconceptions, and respond with questions, redirection, or extensions to deepen their understanding.

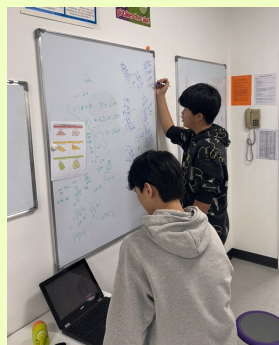
In my experience, there were two challenges to using VNPS. The first was the transition time from the boards back to seated work, which was manageable but noticeable. The second was group dynamics — at times, high-achieving students tended to dominate in their respective group.

Overall, however, I found that using VNPS transformed my lessons into active, visible, and collaborative problem-solving sessions.

A few representative photographs are shown below. They capture very well the mood and atmosphere of these classes.



(a)

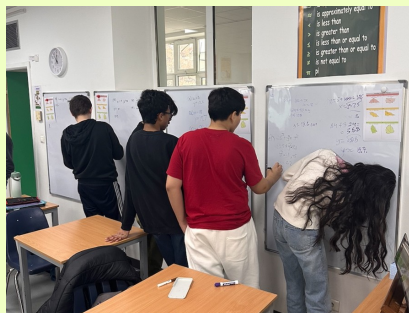


(b)

Figure 1



(a)



(b)

Figure 2

About the author

Shiv Gaur has been teaching mathematics at Tashkent International School in Uzbekistan for the past five years, where he currently serves as Head of Secondary Mathematics. Since beginning his teaching career in 2001, he has taught in India, Indonesia, and Kazakhstan, working with the Indian, IGCSE, and IB curricula. Outside the classroom, Shiv enjoys origami and magic, and stays active through yoga, badminton and table tennis. He may be contacted at. shivgaur@gmail.com

Generating the First 100 Squares from the First 25

Article contributed by Hervé Svoboda

Introduction

As a mental calculation enthusiast, I once heard Dr. Arthur Benjamin — the famous mathemagician — say in an interview that the only thing he knows by heart is the list of the first 100 squares.

One day I thought it might be a good idea to learn those 100 squares myself.

So I started by writing down the list of the first 100 squares, and then I asked myself: *“How am I going to memorize all of these?”*

Faced with the complexity of the task, I decided to take a closer look at the

results to see whether, instead of memorizing them, I could find a quick way to reconstruct them.

A first observation

Looking at my list, I noticed something interesting:

$$26^2 = 24^2 + 100,$$

$$27^2 = 23^2 + 200,$$

$$28^2 = 22^2 + 300, \quad \text{etc.}$$

Which well-known algebraic formula allows us to generalize the previous equalities?

I then realized that if I knew the first 25 squares by heart, I could deduce the next 25 quite easily.

In this article, I propose a method to quickly determine the first 100 squares, assuming we already know the first 25.

Algorithm for calculating squares of numbers from 26 to 75

If we wish to compute the square of a number between 26 and 75, we can use the following algorithm:

- Compute the distance A between the number and 25.
- Compute the distance B between the number and 50.

- The square of the number is then: $100 \times A + B^2$.

Since we already know by heart the squares of numbers less than or equal to 25, we can easily calculate B^2 .

Example

Let us apply this algorithm to 37:

- $A = 37 - 25 = 12$ and $B = 50 - 37 = 13$
- Hence: $37^2 = 100 \times 12 + 13^2 = 1200 + 169 = 1369$.

Questions for the reader

1. Compute in the same way 64^2 .
2. Explain why this algorithm indeed yields the correct square of the given number.

Algorithm for calculating squares of numbers from 76 to 100

If we wish to compute the square of a number between 76 and 100, the algorithm is as follows:

- Compute the distance C between the number and 50.
- Compute the distance D between the number and 100.
- The square of the number is then: $200 \times C + D^2$.

Example

Let us apply this algorithm to 86:

- $C = 86 - 50 = 36$ and $D = 100 - 86 = 14$
- Hence, $86^2 = 200 \times 36 + 14^2 = 7200 + 196 = 7396$.

Questions for the reader

1. Explain why this algorithm indeed yields the correct square of the given number.
2. What happens if we apply our algorithm to numbers between 101 and 125?
3. A nice challenge: Can you devise a similar algorithm to compute the squares of numbers between 126 and 175, assuming we only know the squares of the first 25 integers?

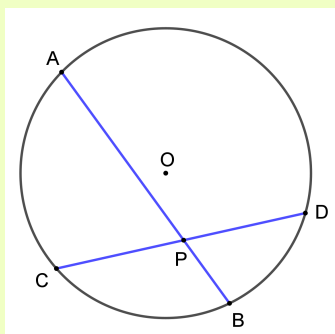
About the author

Hervé is a senior secondary school teacher (agrégé de mathématiques) and has been teaching since 2010 at Lycée Amiral de Grasse in southeastern France. He is passionate about visual mathematics, “proofs without words,” and mental calculation, which he integrates into his teaching and mathematical explorations. His work focuses on making abstract concepts more accessible through imagery and intuition. He may be contacted at svoboda.herve@gmail.com.

The cosine rule — a geometrical proof

In this note, we describe a way of proving the cosine rule using the chord theorem of circle geometry. En route, we encounter (unexpectedly) two different proofs of the venerable Pythagorean theorem.

The chord theorem states the following:



The intersecting chords theorem

If AB and CD are two chords of a circle intersecting at a point P , then

$$PA \times PB = PC \times PD.$$

Figure 3

The following needs to be highlighted.

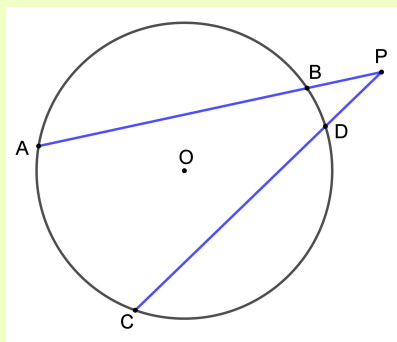


Figure 4

If AB and CD meet at P *outside* the circle, as shown, the statement remains true; we still have

$$PA \times PB = PC \times PD.$$

In this configuration the line segments PBA and PDC are referred to as *secants*.

A special case of the above is when one of the secants becomes a tangent to the circle, e.g., when the points C, D coincide. Such cases, when two otherwise distinct points coincide, are often called 'degenerate cases' in mathematics.

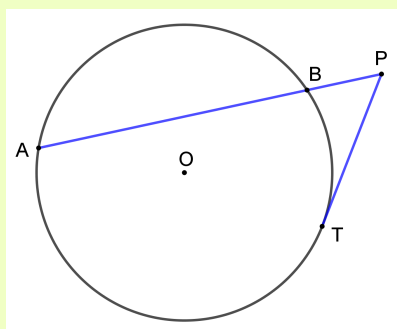


Figure 5

If PBA is a secant and PT is a tangent of the circle, then

$$PA \times PB = PT^2.$$

Now let's see how the chord theorem leads to a proof of the Pythagorean theorem.

Obviously, this is meaningful to do only if the chord theorem can be proved independently of the Pythagorean theorem. Fortunately for us this is the case; the chord theorem does have an independent proof, and it only needs results on triangle similarity. In the first two figures, join AC and BD , and use the fact that $\triangle PAC \sim \triangle PDB$. In the case of tangency, join AT and BT , and use the fact that $\triangle PAT \sim \triangle PTB$. Please complete the proof on your own.

For the proof of the Pythagorean theorem we proceed as follows.

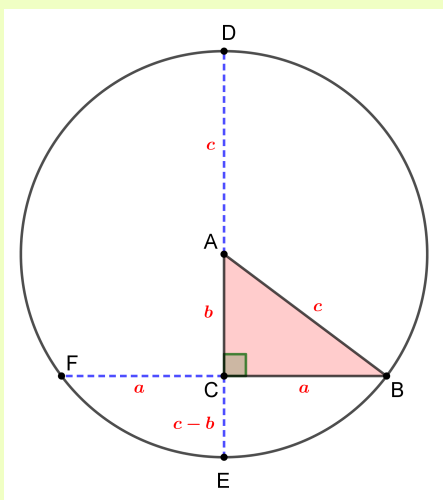


Figure 6

$\triangle ABC$ is right-angled at C .

We must prove that

$$c^2 = a^2 + b^2.$$

Draw a circle with centre A , radius c . Extend AC in both directions to form diameter DE ; then $AD = AE = c$, so $CD = c + b$, $CE = c - b$.

Extend BC to meet the circle again at F ; then $CF = CB = a$.

Now we invoke the chord theorem for chords BF and DE which intersect

at C. We get, $BC \times CF = FC \times EC$,

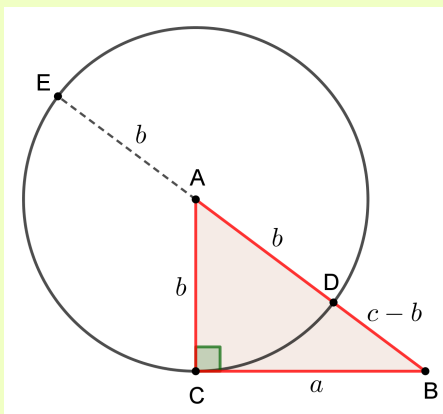
$$\therefore a \cdot a = (c + b) \cdot (c - b),$$

$$\therefore a^2 = c^2 - b^2.$$

Hence, $c^2 = a^2 + b^2$, as required. ■

As you can see, it is a very short proof!

Here's another proof of the Pythagorean theorem, also based on the chord theorem. It is if anything even simpler than the above!



$\triangle ABC$ is right-angled at C.

Draw a circle with centre A, radius b . Let AB meet the circle at D , and let BA extended meet the circle at E ; then DE is a diameter, and $BD = c - b$, $BE = c + b$.

Now we invoke the chord theorem (tangent form).

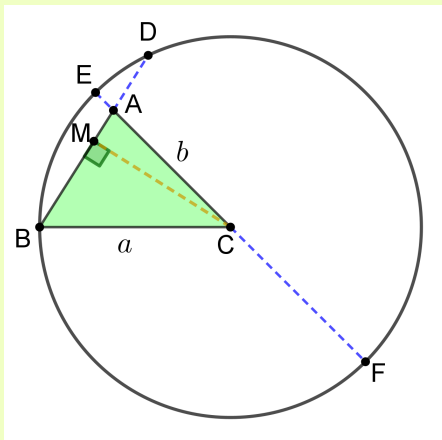
Figure 7

We have, $BC^2 = BD \times BE$, so $a^2 = (c - b) \cdot (c + b)$. This yields,

$$a^2 = c^2 - b^2, \quad \therefore c^2 = a^2 + b^2,$$

as required. ■

We move on to the proof of the cosine rule. It turns out to be a slanted form of the above proof. Here it is.



We are given $\triangle ABC$. We assume for convenience that it is acute-angled, and $a > b$.

Draw a circle with centre C , radius a ; it passes through B . Extend AC in both directions to form diameter EF ; then $CE = CF = a$, hence $AE = a - b$ and $AF = a + b$. Extend BA to meet the circle again at D . Draw $CM \perp AB$.

Invoking the chord theorem for BD and EF , which intersect at A , we get $BA \cdot AD = EA \cdot AF$, hence

$$c \cdot (DM - AM) = (a - b) \cdot (a + b).$$

Next, $DM = BM$ (since $CM \perp BD$), so $DM = AB - AM = c - b \cos A$,

$$\therefore DM - AM = c - 2b \cos A,$$

$$\therefore c(c - 2b \cos A) = (a - b)(a + b),$$

$$\therefore c^2 - 2bc \cos A = a^2 - b^2,$$

$$\therefore a^2 = b^2 + c^2 - 2bc \cos A.$$

We have arrived at the statement of the cosine rule. ■

Remark 1

Note that we have drawn the figure under two assumptions:

- (i) We have assumed that $\triangle ABC$ is acute-angled.
- (ii) We have assumed that $a > b$.

What happens if we relax either of these two assumptions? We will leave this question for the reader to resolve.

Remark 2

The intersecting chords theorem is a humble but mighty result. We will meet it again under various guises.

Article contributed by Shailesh Shirali

How Florence Nightingale Changed Data Visualization Forever

Adapted (and summarized) from: <https://www.scientificamerican.com/article/how-florence-nightingale-changed-data-visualization-forever/>. All images are from this article.

Author: RJ Andrews

Florence Nightingale's revolutionary impact on data visualization is a story of fierce determination, creative innovation, and transformative public health advocacy. Known widely as the "Lady with the Lamp" for her night rounds with a lantern during the Crimean War, Nightingale returned home in 1856 deeply disturbed by the dire conditions she had witnessed in British Army hospitals. Thousands of soldiers succumbed not primarily to battle wounds, but to preventable diseases, aggravated by unsanitary

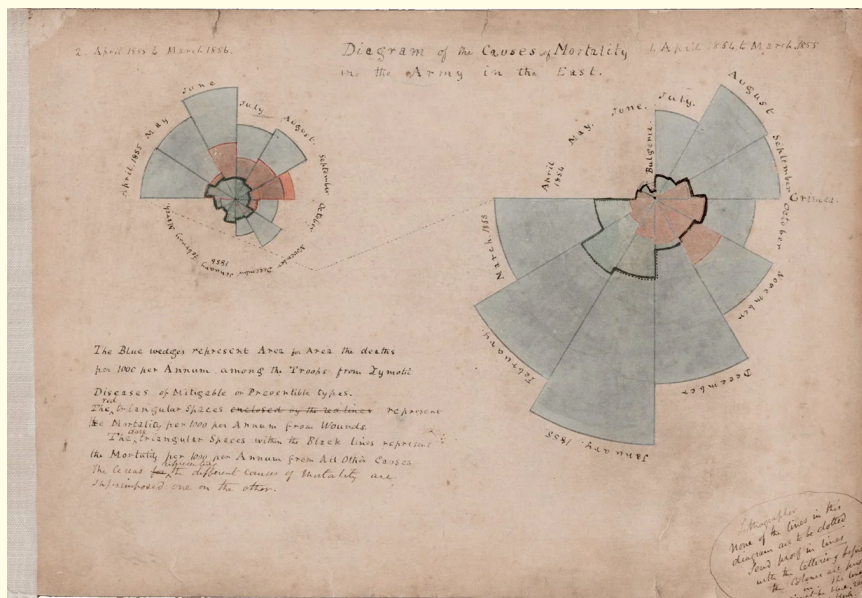
wards, overcrowding, and appalling medical practices. For Nightingale, the real “horrors of war” stemmed as much from administrative neglect as from enemy bullets.



Figure 8: Florence Nightingale

Upon her return to London, she resolved to fight for comprehensive reform. Nightingale recognized that to change entrenched attitudes among generals, medical officers, and Parliament, she would need more than logic — she would need persuasion. At the time, many in charge believed deaths from disease were inevitable, dismissing suggestions that better sanitation would save lives. To make matters worse, the fragmented and inconsistent army records made it nearly impossible to grasp the true causes of death. Nightingale’s quantitative mind was uniquely suited to this challenge.

Faced with poor data literacy among policy-makers — including her prime target, Queen Victoria — she relied on a team of statisticians, scientists, and statesmen to gather and clarify the data. The campaign's central strategy was to tell stories with data—using visuals that could excite and engage non-experts. Her methods stood in stark contrast to the dry, complex statistical tables and thick analytical books preferred by other reformers. Instead, Nightingale's graphics were designed for clarity, impact, and accessibility, combining punchy diagrams with witty prose in slim folios that attracted attention.



Through these visuals, Nightingale communicated the urgency of reform and the reality of preventable death. She ingeniously constructed comparisons to lay bare the consequences of administrative neglect — contrasting mortality rates among soldiers against those of civilians, and showing how peacetime army deaths exceeded civilian deaths for similar age groups. These compelling stories, told through charts, illuminated the failures of military sanitation and made it impossible for the authorities to ignore the evidence.

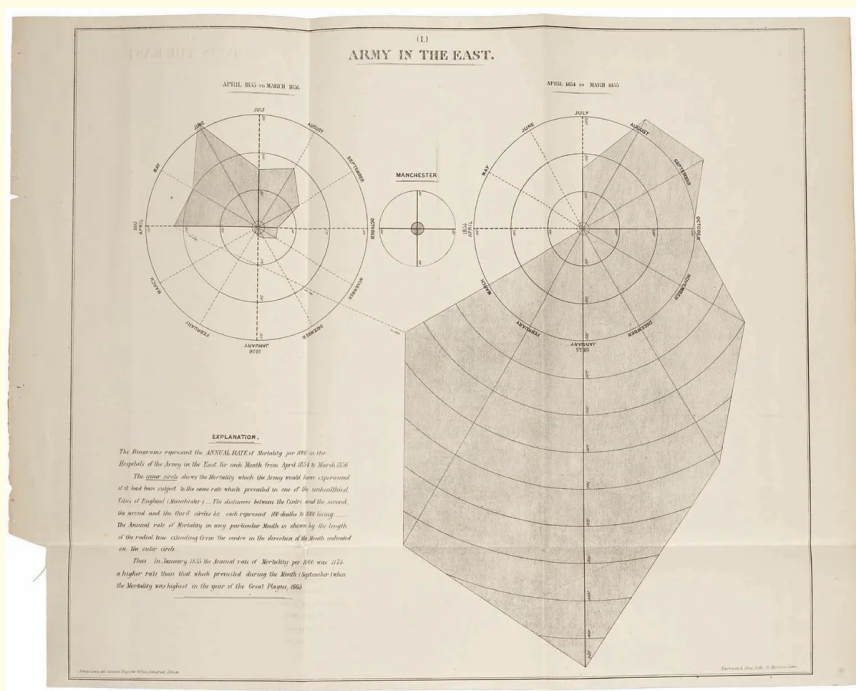


Figure 10: Another famous graphical depiction from Florence Nightingale

Nightingale's diagrams captured national attention. Within months, Parliament debated army sanitary conditions, resulting in significant reforms and the creation of dedicated commissions on health and data collection. Her approach worked; new data gathering systems — soon celebrated as the best in Europe — proved that sanitary reforms drastically reduced mortality rates among soldiers. Nightingale marked this achievement with a final Crimean War diagram in 1863, visually representing the drop in deaths from preventable diseases.

Her greatest legacy unfolded in the wider sphere of civilian public health. The British Public Health Act of 1875, which owed much to Nightingale's efforts, set the foundation for modern sanitation: regulated sewers, clean running water, and improved building codes. This set off a century-long transformation, as improved hygiene and health infrastructure, coupled with advances such as vaccines and artificial fertilizers, nearly doubled average human life expectancy worldwide.

The article reveals stunning details of Nightingale's creative process. Surviving drafts and correspondence show her leadership in refining graphical information — not just overseeing, but actively managing every aspect from data wrangling to design, funding, and distribution. She worked with the eminent statistician William Farr and government clerks to produce various chart types: bar charts, area diagrams, and her now-iconic “rose” or polar area diagram. These diagrams visually dramatized the causes and rates of death—making abstract statistics powerfully real and actionable.

Nightingale's best-known graphical innovation is the set of three diagrams originally crafted for Queen Victoria and later publicized as part of a rebuttal to critics. This narrative sequence — a model for modern data storytelling—progressed from exposing the extent of the problem (excess

deaths), to revealing causes (preventable disease), and finally spotlighting the effective solutions (sanitary reforms). Her diagrams varied in style and encoding, sometimes even correcting visual exaggerations from earlier versions, as Nightingale and her team continually refined their approach for maximum interpretability and impact.

The significance of Nightingale's work lies not only in its historical effect but in its methodological foresight. She understood that effective communication of statistics requires narrative and visual engagement—anticipating the central role of data visualization in today's public debates over health, policy, and science. Nightingale's fusion of statistical rigour with visual storytelling helped create the very language of modern information graphics, shaping both societal understanding and policy decisions for generations to come.

This article thus positions Florence Nightingale as much more than a pioneering nurse — she emerges as a foundational innovator in data visualization, whose methods remain vital, relevant, and inspiring to all who seek to use data for social good. Her work demonstrates that the true power of data lies in its ability to tell urgent, compelling stories that can move minds and change lives.

(This is a summary of the original Scientific American article, produced using the LLM Perplexity.)

The Big Wide World of Data

Visualisation – 6

As stated in earlier issues, this column aims to present a running series of case studies and/or short articles on the theme of data visualisation. We shall present a few striking examples of this in each issue (with little or no commentary). The graphs shown here are worth studying closely for the techniques used.

How to fight climate change without the US

From the article “How to fight climate change without the US: a guide to global action” by Jeff Tollefson, available at <https://www.nature.com/articles/d41586-025-03571-x>

The article discusses the implications of the US’s planned exit from the Paris climate treaty for global efforts to fight climate change.

TRUMP'S CLIMATE LEGACY

US greenhouse-gas (GHG) emissions are projected to be higher under President Donald Trump than they would have been under the policies of former president Joe Biden. This forecast takes into account the impact of Trump's policies and of federal legislation to cut government support for clean-energy programmes.

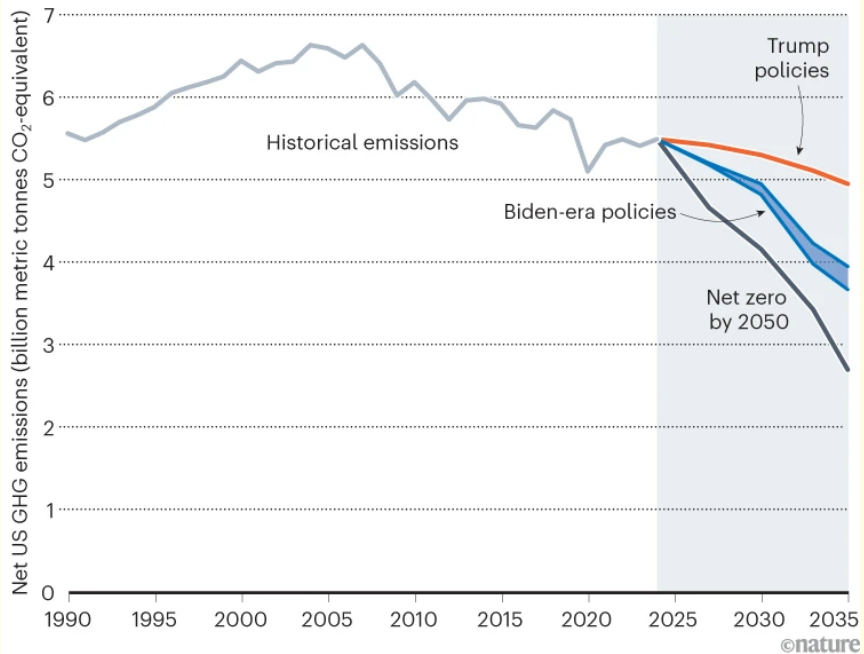


Figure 11: Trump's climate legacy. Source: <https://www.nature.com/articles/d41586-025-03571-x>

Despite the US being the world's second biggest greenhouse-gas emitter, other major actors like China, India, the European Union, Brazil, and US states and cities are taking significant initiatives for climate action. The

upcoming COP30 conference in Belém, Brazil, is seen as a crucial moment to lock down new commitments for limiting dangerous climate change, with Trump administration not expected to participate meaningfully.

Will China take the lead on climate?

- China is the world's largest emitter and has driven 90% of emission growth since 2015, but is also the global leader in adopting clean energy and manufacturing renewable equipment.
- President Xi Jinping announced goals to reduce greenhouse-gas emissions by 7–10% from peak levels by 2035; there's debate about whether this is ambitious enough but it is seen as compatible with the Paris treaty.
- China faces challenges such as continued building of coal-fired power plants and prioritizing economic growth over environmental goals post-COVID-19.
- China's leadership at COP30 may focus on wielding influence among developing countries and supporting financing for tropical forest preservation.

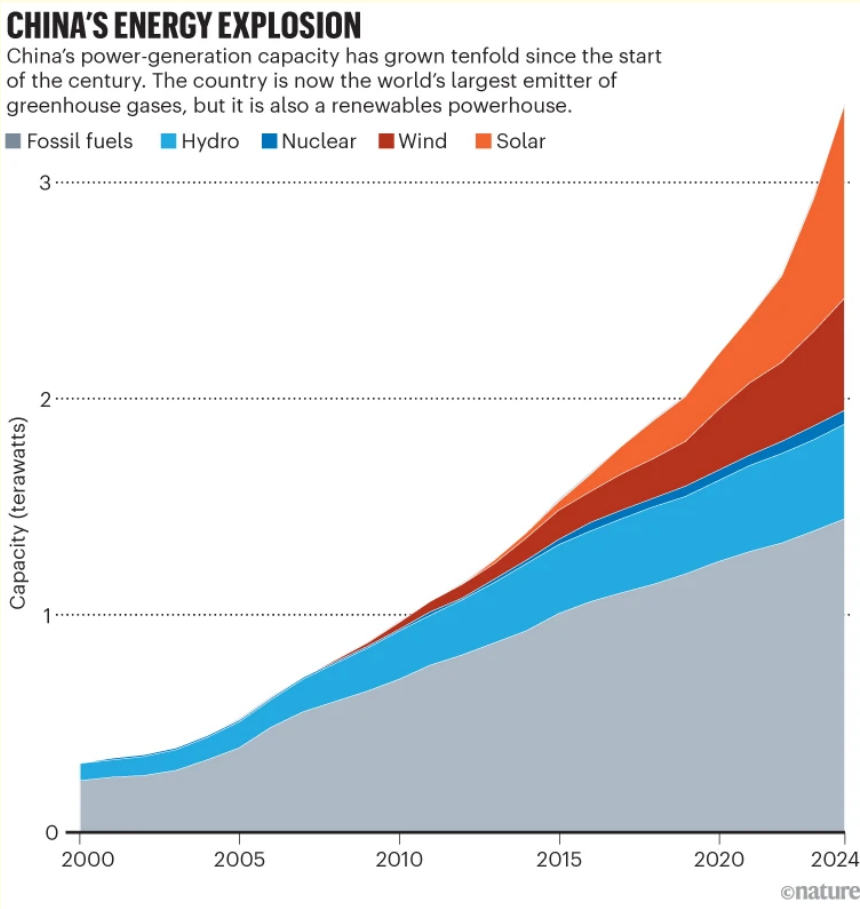


Figure 12: China's energy explosion

Brazil’s net-zero potential

BRAZIL YO-YOS ON DEFORESTATION

Deforestation and accompanying changes in land use are the main sources of greenhouse-gas emissions in Brazil, which wax and wane depending on who is in power and what policies the government adopts.

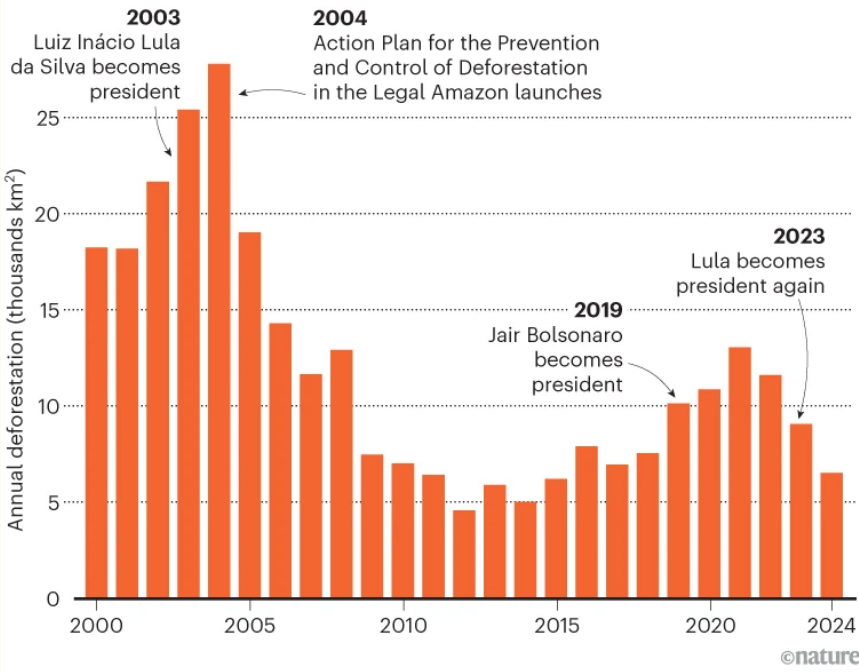


Figure 13: Brazil’s net-zero potential

- As COP30 host, Brazil aims to facilitate negotiations and lead by example, with much at stake due to climate impacts like devastating floods.

- Tackling deforestation in the Amazon is highlighted as Brazil's key opportunity to reduce emissions, with recent investment in funds to preserve forests.
- Despite progress, some government sectors continue to pursue projects that may increase emissions (like highway or oil drilling projects), and Brazil's latest climate targets are criticized as not ambitious enough.
- However, Brazil's potential for shifting to renewables (wind, solar) and halting deforestation is seen as achievable with relatively low economic downside.

India motors on climate goals

- India is moving forward on its climate commitments, achieving 50% non-fossil-fuel electricity generation five years ahead of schedule.
- India's push for renewables is seen as driven by energy security needs and self-reliant economic growth, not just global agreements.
- Coal remains central to India's energy plan, justified by its low per capita and historical emissions, though a recent surge in renewable energy is slowly replacing coal-powered energy.
- US policies could indirectly affect India's climate trajectory, particularly via disruptions to global trade and economic pressures, which may shift priorities away from decarbonization.

People are having fewer babies: Is it really the end of the world?

From the article “People are having fewer babies: Is it really the end of the world?” by Lynne Peeples, available at <https://www.nature.com/articles/d41586-025-02615-6>

The article begins by illustrating the dramatic drop in fertility rates around the globe, from countries like Mexico — where the average number of children per woman fell from seven in 1970 to less than two by 2023 — to similar trends observed in most nations. Experts note that more than three-quarters of countries will face below-replacement birth rates by 2050, raising widespread concern. Although some fear this “baby bust” could lead to slumps in innovation, productivity, and support for ageing populations, others argue that the decline’s potential impact depends on how societies prepare and adapt.

What the data say

The article now traces global fertility trends from the mid-20th century, recounting how dire warnings of overpopulation have given way to new anxieties about shrinking populations. While the world’s average fertility rate has dropped to 2.2, about half of all countries are now below the 2.1 threshold needed for population stability. South Korea provides a stark example, with fertility dropping from 4.5 in 1970 to 0.75 in 2024. Despite some regional differences — sub-Saharan Africa retains high birth rates — demographers broadly agree the global population will peak in the coming

decades before beginning its first sustained contraction since the 1300s.

DECLINING FERTILITY

Most countries throughout the world have seen a decline in total fertility rate — the average number of children a woman might expect to have during her reproductive years. In many regions, the rate has dipped below the threshold needed to maintain a population (generally around 2.1).

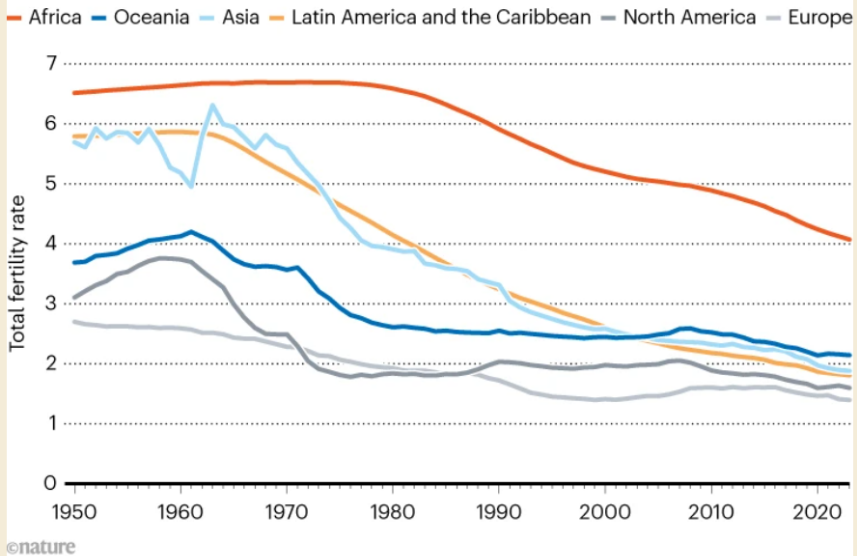


Figure 14: Declining birth rates

What's driving the decline?

The article moves on to explore the complex causes behind falling fertility rates. These include expanded access to contraception, higher levels of education and career opportunities for women, evolving cultural norms,

increased financial stress, and shifting attitudes toward parenting and partnership. In some cases, such as Iran, rapid policy changes have triggered dramatic fertility shifts. Additionally, economic pressures, rising housing costs, changing parenting cultures, environmental anxieties, and even declining sperm counts are cited as driving forces. The underlying crisis is described as broken systems and institutions preventing people from having the number of children they desire.

Countering the crash

The article now looks at the varied responses governments have tried, from financial incentives like baby bonuses to subsidies for childcare and housing. While such measures offer modest, often short-lived increases in birth rates, researchers stress that no single policy will restore high fertility. More promising are progressive schemes granting parental leave and valuing care work, especially where men share care-giving responsibilities. The article argues, however, that adaptation to new demographic realities deserves equal if not greater emphasis: slow but steady gains in fertility can provide time to prepare, but policies must support broader social resilience.

Adapting to a new reality

In the concluding section the article lays out strategies to respond to shrinking populations. These include strengthening the care workforce, raising retirement ages, and reforming pension systems to accommodate longer lifespans and growing numbers of elderly people. Immigration is presented as a vital tool for mediating labor shortages, although its political sensitivity and impact on sending countries are acknowledged. Experts rec-

commend exploring less obvious avenues — like eradicating child poverty and creating conditions for hope and well-being — since happier, healthier people are more likely to have the children they want. The article ends on a note of cautious optimism: declining fertility, if met with appropriate adaptation and investments in health and education, can yield benefits for individuals and society overall. The real disaster, the author concludes, is failing to adapt.

Teaching data handling in a collaborative way using the 'Surveyypura' tool

Article contributed by Mukta Gundi & Adithya Pradyumna

The importance of learning 'data handling' in today's world

There is now a growing emphasis on quantitative data analysis as part of various professions. This includes making decisions about processes and systems in social welfare, health programmes or the transportation and logistics sector etc. The Government as well as the private sector are dependent on inferences based on data analysis. Additionally, data visualisations based on quantitative data analysis are often used to communicate with the public on important topics, such as we saw during the COVID-19

pandemic. Therefore, learning skills related to data generation, analysis and interpretation are important in today's world.

The syllabus on data-handling for classes 6, 7 and 8

The National Council of Educational Research and Training (NCERT) syllabus for Mathematics includes the topic of data handling for grades 6, 7 and 8. The concepts are introduced in an incremental way in these syllabi. For class 6, it starts with an introduction to quantitative data; the process of data collection, and organization (tabulation), and data analysis including visual representation of the data. For class 7, there are additional components such as calculating mean, median, and mode, and introduction to percentages. The concept of probability with more advanced problems along with data visualisation get discussed in class 8 syllabus. While these concepts are being introduced, it is important to ensure that data handling does not remain a mechanical procedure. School teachers should help students think about questions such as 'what is quantitative data, how is it connected to the world, why observation is an important skill as part of data collection and how to record data'.

The need for experiential learning of 'data handling'

The *Ganita Prakash* [reprint 2025-26 version], which is the NCERT textbook for Mathematics, provides ideas for teaching these concepts in an engaging way. The textbook also encourages teachers to bring examples from the school context (such as, the number of students absent across different classes, tabulating frequency of the favourite games of the students in class) so that students can better relate to numerical data and quantitative concepts.

The National Education Policy of 2020 states that "pedagogy must evolve to make education more experiential, holistic, integrated, inquiry-driven, discovery-oriented, learner-centred, discussion-based, flexible, and, of course, enjoyable." Indeed, the new NCERT textbooks try to embody these principles. We believe that the Surveyypura tool, which is a pedagogical tool that we have developed, can be useful to teach data handling through a fun and immersive data collection experience.

About Surveyypura: A simulation-based pedagogical tool

The Surveyypura tool consists of a few components. The first is a large illustration of the imaginary village named 'Surveyypura'. It shows a total 155 houses and the other amenities that are typically seen in an Indian village, such as a primary school, kirana shop, fields. (Figure 1). This illustration is meant to be printed on canvas and is of the size 16 ft by 4 ft. It can be kept in the centre of the classroom (either on the floor or on tables) when in use. The village is colourful, giving a sense of a vibrant setting, immediately attractive to students, who can see different types of houses, paved and unpaved roads, people going about their work, livestock etc. There are three types of houses in this village - kachha, pucca and semi-pucca based on the types of roofing and housing materials (Figure 2). Each house has been assigned a three-digit number [Household identification number or HHID], from 001 to 155, printed on the respective household in sequential order.



Figure 15: Illustration of the fictional Surveyypura village

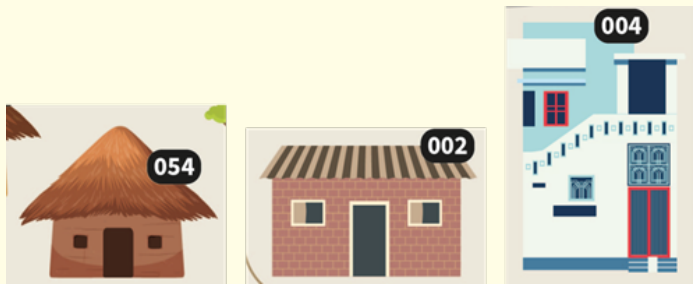


Figure 16: Examples of different house types on the map [L to R]: kachha, semi-pucca and pucca

The second component of this tool is a set of 155 data cards — one card per household. Each card has some demographic data for that particular household. The data cards are brown in colour, showing the HHID at the top which corresponds with a particular house in the village on which it is placed (Figure 3). Table 1 provides the list of five questions for which data has been provided on the cards.

- 1a. What is the total number of people living in the household?
- 1b. How many household members are under the age of 5 years?
- 1c. What is the caste category of the head of the household?
- 1d. What is the total land owned by the household? (in acres)
- 1e. Does the household have a toilet?

Table 4: Questions for which data has been provided for each household (see Figure 2 for how the responses look on a data card)

HHID - 001

(1a) 4

(1b) 0

(1c) General category

(1d) 1 acre

(1e) Yes

Figure 17: Sample data card, back and front

How to use Surveyypura for the classes 6 to 8

As the concepts related to data handling are being introduced in an incremental way in classes 6, 7 and 8, this tool can be used at each stage of learning. While the canvas is necessary for the activities across these classes, the data cards may not be necessary for class 6 students. Teachers will be able to come up with additional ideas on how this tool can be used in accordance with the learning objectives of their respective class syllabi.

Step 1: Arranging the tool together

Teachers can include the students in arranging the tool at the centre of the classroom (Figure 4). This will ensure collaboration at the beginning of the class activities.



Figure 18: Students arranging the tool and exploring Surveyypura village

Step 2: "Visiting" Surveyypura

It would be helpful to first let the students observe the Surveyypura village to generate curiosity and channelise their excitement in a more constructive way. Teachers can ask them to identify and note down different characteristics of the village by asking questions such as "which side of the village do you see more two-storey or pucca houses?" or "where is the school?".

Step 3: Understanding more about Surveyypura village through data collection, analysis and interpretation

Below we describe some activities which can be facilitated over a few lectures to introduce and discuss the concepts in an engaging manner. Students can be allocated into groups — each with 4-5 students. A certain range of household ids can be allocated to each group. For instance, group 1 can cover HHIDs 001 to 020.

Step 3.1 The teacher says to the class: "Let's try to understand more about this village Surveyypura. Can you observe and tell me how many houses there are in this village?". The students are given some time to count the total number of houses in the village.

Step 3.2 The teacher then asks, "Can you now tell me the different types of houses in Surveyypura?". The students will be able to recognise that the houses look different from each other and have different features. The teacher can help the students appreciate the three types of houses: kaccha, semi-pucca and pucca and discuss why it has been classified this way.

Step 3.3 The teacher says to the class: "Let us now count how many houses of each type there are. For counting, we first need to prepare a data collection form. In this form, you will be noting down what you observe about the type of house". The teacher can then show a data collection form as shown in Figure 5 and Ask them to copy it and go to the allocated houses and note down the house type.

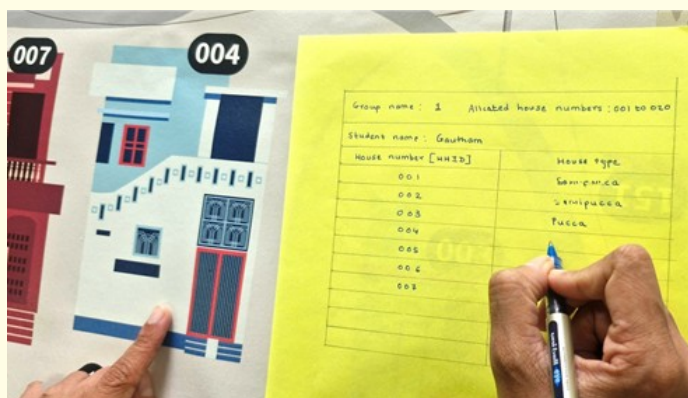


Figure 19: Noting down data through observation during the survey on house types

Step 3.4 The teacher can then ask the students to tabulate the data: "Now we want to know how many of each type of houses there are in the houses allocated to you. For this, create a frequency table or a pictograph for the house types based on the data you have collected.". An example of the tabulation results is shown in Figure 6.

House type	Tally	Total
Kachha		0
Semipucca		4
Pucca		7
		20

Figure 20: Tabulating the results on house types using tally marks

The below-mentioned group activities using the data cards along with the Surveyypura map would be more relevant for Grade 7 and Grade 8. Students can work in groups to place the data cards on the respective houses by matching the HHID mentioned on the card with that of the house for their allocated HHIDs.

Step 3.5 The teacher informs the next step, "For the houses allocated to your group, we want to know a few things: what is the average number of people per household across the households, what the average land owned by households is, what proportion of houses have toilets, and what proportion of houses are kachha? We can get this data by going to those houses. Some of this information is in the data card on the house, and we can know about what type of house directly by seeing the house. Each data card has answers for five questions. But for what we want to know, we only need answers for three questions, which are: 1a [What is the total number of members in the household?], 1d [What is the total land owned by the household?], and 1e [Does the household have a toilet?]. You will need to read the data card on each of the allocated houses and note down the answers provided. Of course, you will first prepare a data collection

form, which will help you note down all the necessary information.". The teacher can show a data collection form like what is shown in Figure 7.

Step 3.6 The teacher then says, "Hopefully all of you have written it down correctly! Now as a group, together we calculate the average number of people living per household in the allocated houses and the average land owned per household in acres in the allocated houses". Similarly, the median and mode can be calculated using this data.

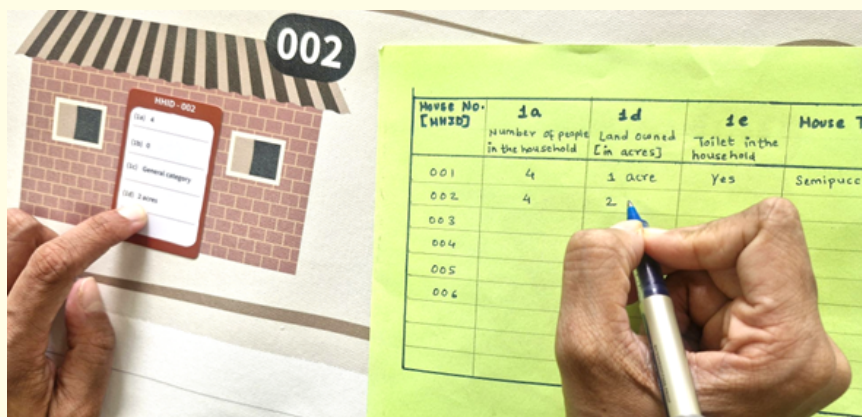


Figure 21: Noting down data from the household data cards and from observation

Step 3.7 Students can now be encouraged to individually tabulate their results [a sample tabulation is shown in Figure 8]. They can be encouraged to calculate the percentage of the allocated households having a toilet, and the percentage of kachha, semi-pucca and pucca houses.

Step 3.8 After tabulating the data, the teacher can introduce the concept of data visualisation using pie charts. Students can be encouraged to draw a bar chart and pie chart also for the percentage of kachha, semi-pucca and pucca houses.

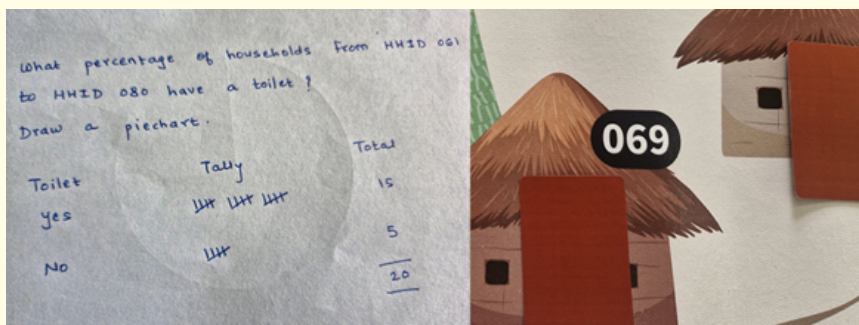


Figure 22: Using tally marks to tabulate the frequency of houses in the sample that have toilets

Concluding remarks

Surveyapura can provide immersive experience of collecting, entering, analysing and interpreting data. Our experience of using this tool at the university to teach survey research methods course revealed that this tool encourages students to work step-by-step and collaboratively while also being physically active in the process of data handling in a fun and engaging way.

This tool can also facilitate cross-learning as students can connect mathematical concepts with social studies [for e.g. discussion around why does the group with maximum kachha houses have the least proportion of toi-

lets as compared to other groups with more number of pucca houses?]. We look forward to your feedback!

More information about the tool

- All necessary links and contact information: <https://azimpremjiuniversity.edu.in/pedagogical-tool/surveypura-simulation>
- Video explaining the features of the tool: <https://www.youtube.com/watch?v=xSUu75YktUY&t=1s>
- Contact/queries - write to us: adithya.pradyumna@apu.edu.in and mukta.gundi@apu.edu.in

Further reading

- [1] Ganita Prakash: Class 6. 2025-26.
<https://ncert.nic.in/textbook/pdf/fegpl104.pdf>
- [2] Ganita Prakash: Class 7. 2024-25.
<https://ncert.nic.in/textbook/pdf/gemh103.pdf>
- [3] Mathematics: Class 8. 2024-25.
<https://ncert.nic.in/textbook/pdf/hemh104.pdf>
- [4] Pradyumna, A., Gundi, M. Reflections on the design and application of 'Surveypura': a simulation-based pedagogical tool for quantitative research methods in public health and social sciences. Adv Simul 9, 2 (2024). <https://doi.org/10.1186/s41077-023-00275-y>

[5] National Education Policy 2020. Government of India.

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Disclaimer: The views expressed in this article are those of the authors and do not necessarily represent those of Azim Premji University.

Tariff and Impact on Foreign Exchange Rate

Article contributed by Prabha Chaitanya

Tariffs have been in news for several months now. The United States took the world by storm when President Donald Trump announced “Liberation Day” on 2nd April, 2025, a day when import duties were levied on several countries.

What is Globalization? Globalization is a process by which there is free flow and movement of goods, services and people across geographical borders. Some regions are naturally rich in resources, knowledge or competencies, and are in a position to cater to global demands. Globalization, if approached in the right perspective, can enable a single global economy. It implies greater inter-dependency and inter-connectedness between nations. But if such inter-dependency comes under political duress, vulnerable nations can be put to a severe test through economic tools (e.g., tariffs) that challenge free flow of goods, services and people.

What is a Tariff? A tariff is charged on goods that are imported into a country. As an example: if there is no tariff, and the cost of a good is \$50, with a tariff of 10%, the cost becomes \$55. Therefore, a tariff acts as a deterrent to globalization. It escalates the cost of the good. What are some of its possible implications?

- 1) If the good is a raw material, the cost for the producers within the country will escalate, leading to an increase in the price of a good. If the U.S imports several components or raw materials, we can expect a rise in the general price level within the U.S.
- 2) If it is a final consumption good and there is a tariff, then its cost will rise and consumers within the country may reduce its purchase. The exporting country could be impacted adversely.
- 3) If it is an essential good for the U.S, then an increase in cost will not prevent the domestic consumer from purchasing it. Then exports from say, India to U.S for that essential commodity may not be adversely impacted.

What is Foreign Exchange Rate? Foreign Exchange rate refers to the price of a foreign currency. It refers to the number of units of the domestic currency that exchange for 1 unit of a foreign currency. For example: \$1 = INR 87.50.

If the demand for foreign currency increases and the supply of foreign currency is unchanged, one would see the price of 1 unit of the foreign currency go up. Such an increase in demand for foreign currency can come about due to a variety of reasons, as indicated below.

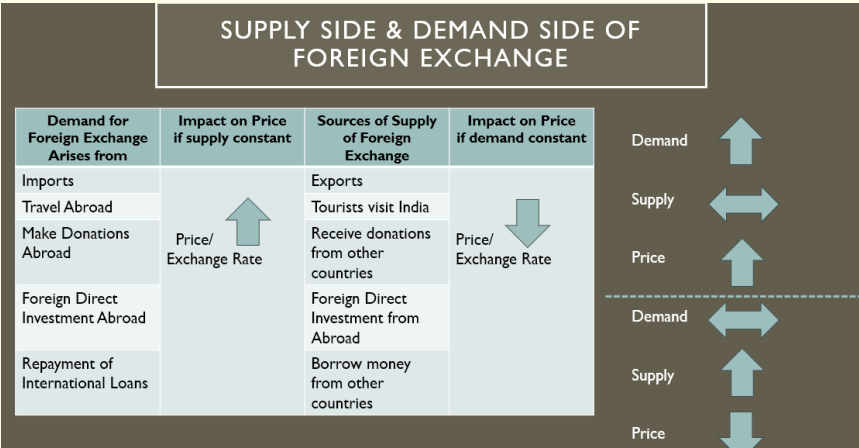


Figure 23

Taking the current scenario, if India’s exports are impacted adversely (as higher tariffs are imposed by the US), then the supply of foreign currency in India reduces. Domestic goods are not competitive in the U.S market as their price has risen due to the levy of a tariff. It pushes up the exchange rate. (termed depreciation of the domestic currency). A fall-out is that importing goods becomes more expensive. For a developing country, which focuses on importing capital goods, it could impact investment and growth. This is shown graphically below:

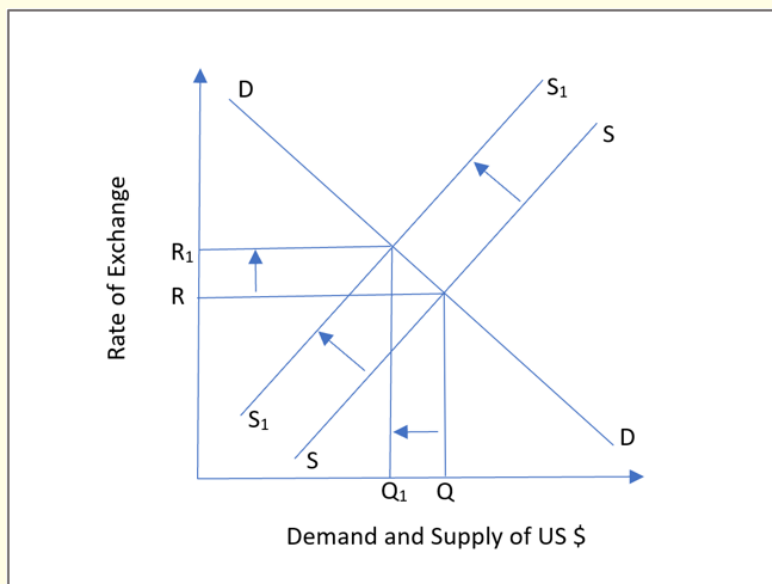


Figure 24

If there is a certain demand for foreign exchange and supply of foreign exchange, an equilibrium rate of exchange is established at R (as an example, let us say $\$1 = \text{INR } 80$). If the supply of foreign exchange reduces, due to a decrease in India's export of goods to the US (in response to tariffs), one can see the supply curve SS shift to S_1S_1 . As can be clearly seen, such a shift in the supply curve, causes the rate of exchange to increase to R_1 (as an example, let us say $\$1 = \text{INR } 85$). Such a rise in the exchange rate implies that for every additional purchase of a dollar, one would have to pay INR 5 more. This would certainly have a dampening effect on the imports (given the thousands of dollars worth of imports India makes).

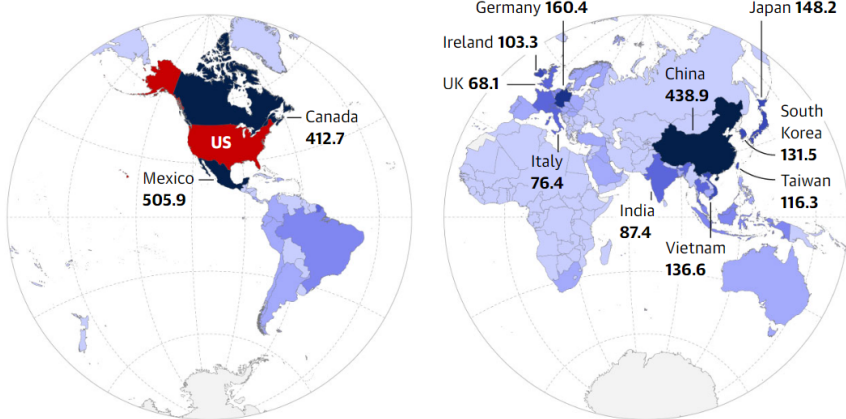
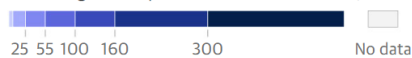
Let us look at certain data and reasons the U.S cites for imposing tariffs.

Balance of Trade Countries trade in goods and services. Balance of Trade refers to the net balance (in monetary terms) after import and export of goods has taken place between two countries.

- Imports > Exports = Deficit Balance of Trade
- Exports > Imports = Surplus Balance of Trade

Countries exporting the most goods to the US

Value of goods exported to the US in 2024, \$bn



Source: US International Trade Administration. Note: top 12 named

Figure 25: Source: <https://www.theguardian.com/business/2025/apr/02/liberation-day-what-is-a-tariff-and-why-they-matter-donald-trump>

The US is the largest goods importer in the world – buying products worth \$3tn in 2023. It also has the largest trade in goods deficit (imports exceed exports) – worth \$1tn.

This is the primary reason cited for tariffs. U.S believes it can earn revenue by levying these tariffs and wants the world to set up manufacturing units in the U.S.

US balance of trade has stood at a deficit of at least \$500bn since 2017

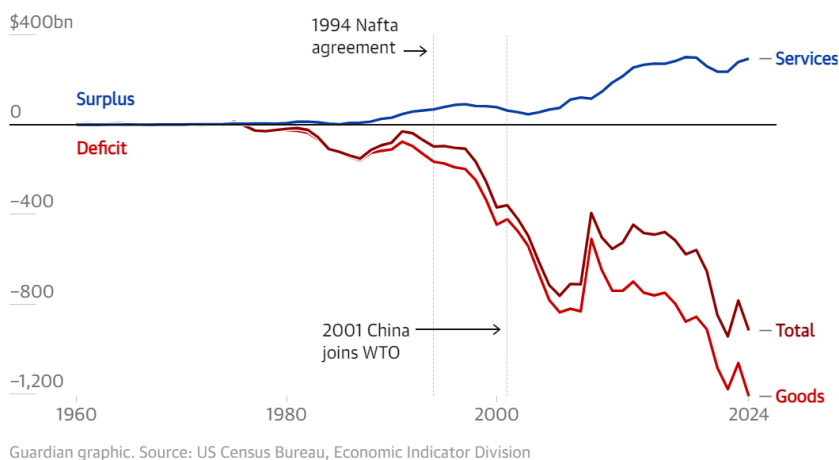


Figure 26: Source: <https://www.theguardian.com/business/2025/apr/02/liberation-day-what-is-a-tariff-and-why-they-matter-donald-trump>

“Growth is expected to slow markedly to 1.6% in 2025 and 1.5% in 2026. This reflects the substantial increase in the effective tariff rate on imports and retaliation from some trading partners, high economic policy uncer-

tainty, a significant slowdown in net immigration, and a sizeable reduction in the federal workforce.”

United States 1

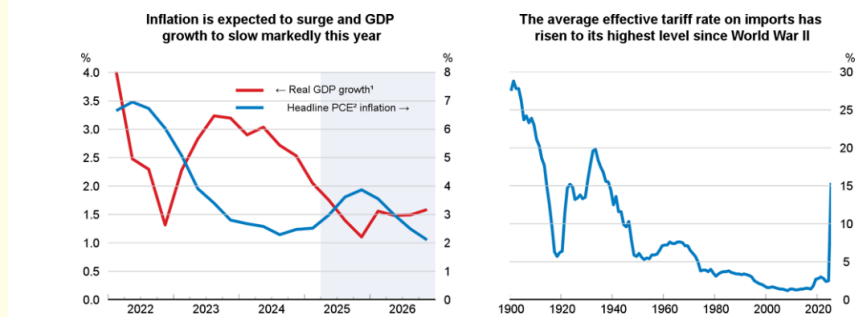


Figure 27: Source: https://www.oecd.org/en/publications/oecd-economic-outlook-volume-2025-issue-1_83363382-en/full-report/united-states_c69a8d2f.html

U.S dollar and INR Exchange Rate

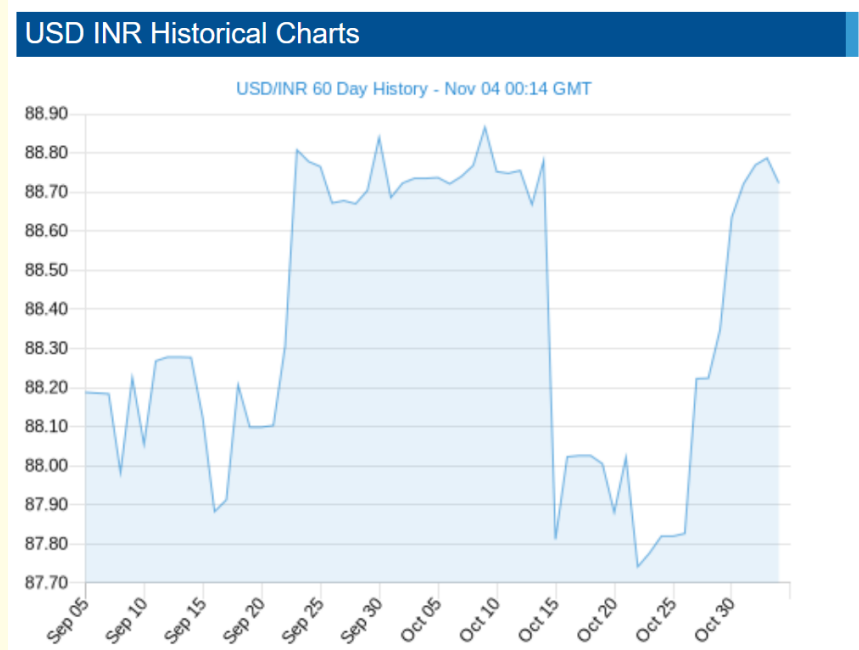


Figure 28: Source: <https://www.exchangerates.org.uk/USD-INR-exchange-rate-history.html>

Tariffs in place against various countries: As we write this, negotiations between the U.S and several countries are either concluded or underway. 50% tariffs on goods from Brazil; 30% tariffs on South Africa; 20% tariffs on Vietnam; 19% tariffs on Indonesia and Philippines and 15% tariffs on Japan and South Korea. The rate that will be charged on Indian goods is set to climb to 50% on 27 August, 2025 (as of 17th August, 2025).

The European Union struck a deal with the US during Trump's window

for negotiations. In late July, the two sides agreed that European goods would face 15% tariffs, including on cars. Under the deal, which awaits approval by all 27 EU members, the trading bloc will charge US firms 0% duty on certain products. A 35% tariff will apply to all Canadian goods over and above the existing duties. Trump has delayed imposing higher tariffs on all Mexican goods for another 90 days, awaiting negotiations. US-China trade negotiations are underway. After raising tariffs on each other's goods initially and then temporarily lowering rates for a 90-day period, they decided to hold onto those rates till 10th November.

Concluding Thoughts: What are retaliatory tariffs? When a country imposes a tariff on imports from a country, in response to a tariff imposed on its exports. India aims to focus on bilateral trade deals with the US to bring down duties and seeks to avoid retaliation on US tariffs. While a slowing down of GDP is predicted in most economies, it is believed also that the world economy has grown to such a size that countries will find a way around America's protectionism (protecting the domestic economy). Countries will start trading 'around America'. 80% of global trade in goods does not directly touch the US, according to research of Boston Consulting Group.

Questions to Ponder

- 1) U.S Balance of Trade deficit was at least \$500 Bn since 2017. Under what circumstances would a tariff on an exporting country, not serve the U.S intention of correcting such a deficit? (refer U.S Balance of Trade graphic)
- 2) From the Balance of Trade graphic, what other options seem possible to correct the Balance of Trade deficit?

3) What kind of a relationship seems to emerge between Real GDP growth and inflation in the U.S from 2022-2026 (from the OECD graphic)?

Further reading

- [1] <https://www.theguardian.com/business/2025/apr/02/liberation-day-what-is-a-tariff-and-why-they-matter-donald-trump>
- [2] <https://www.theguardian.com/business/2025/apr/04/trump-tariffs-in-economic-terms-it-makes-no-sense-at-all>
- [3] https://www.oecd.org/en/publications/oecd-economic-outlook-volume-2025-issue-1_83363382-en/full-report/united-states_c69a8d2f.html
- [4] <https://comtradeplus.un.org/>
- [5] <https://www.exchangerates.org.uk/USD-INR-exchange-rate-history.html>
- [6] <https://www.bbc.com/news/articles/cn93e12rypgo>

About the author

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India's got a glorious math history, now it will have a future

"India's got a glorious math history, now it will have a future. Lodhas getting the best brains." <https://theprint.in/ground-reports/indias-got-a-glorious-math-history-now-it-will-have-a-future-lodhas-getting-the-best-brains/2728301/>. Author: Akanksha Mishra

India is embracing a new chapter in mathematics research with the founding of the Lodha Mathematical Sciences Institute (LMSI) in Mumbai, backed by the Lodha Foundation and led by eminent mathematician Vijaya Kumar Murthy. The institute aims to revive India's vast historical legacy in mathematics — home to luminaries like Aryabhata, Bhaskara II, and Ramanujan — by instituting a space dedicated solely to high-level math-

ematical research and collaboration without the distraction of teaching responsibilities or administrative burdens.[1]

Vision and Foundation

LMSI was conceived in 2024 through a significant philanthropic act by the Lodha family, with Abhishek Lodha, CEO of Lodha Group, as its driving force. The institute intentionally mirrors renowned global centres such as the Institute for Advanced Study (IAS) in Princeton and the Fields Institute in Toronto, aiming to provide a focused environment for pure mathematical research. The absence of Indian citizens as Fields Medal recipients — the most prestigious international mathematics prize — despite considerable talent, is a notable motivation for this initiative.

Academic Leadership and Structure

The scientific advisory panel consists of distinguished mathematicians, including Manjul Bhargava (Princeton), Saurav Chatterjee and Ravi Vakil (Stanford), Alexander Lubotzky (Weizmann Institute), and Kavita Ramanan (Brown). The institute's operation is unique: it does not offer classes or degrees but organizes six-month thematic programs where global scholars assemble to work intensively on particular topics. The first program, focused on arithmetic statistics and led by Bhargava, runs from August to December and is followed by programs led by other prominent mathematicians.

Symposium and Academic Community

LMSI's inaugural symposium brought together forty leading mathematicians from India and abroad for in-depth discussions spanning topics from convex geometry to artificial intelligence and quantum field theory. The emphasis is on Socratic dialogue and the free exchange of ideas, with the ultimate aim of stimulating collaboration and innovation within India. The institute hopes to foster the return and retention of Indian mathematical talent, which traditionally sought opportunities abroad due to the lack of similar research-focused institutions at home.

National Impact and Future Goals

Abhishek Lodha envisions LMSI's success measured not only by academic achievement and faculty quality but by a broader transformation: inspiring a new generation of Indians to pursue mathematics professionally and fostering technological and social progress rooted in mathematical innovation. Plans include delving into applied areas with real-world impact, such as cryptography, artificial intelligence, and traffic modelling, addressing major national technical challenges through foundational mathematics.

The LMSI Approach

LMSI's core philosophy is to provide spaces for both contemplation and collaboration, widely regarded as essential for deep mathematical thought and breakthroughs. The institute aspires to become a nurturing environment where scholars—especially emerging Indian mathematicians—can

flourish and make significant contributions to global mathematics from within India.

In summary, the Lodha Mathematical Sciences Institute is positioned as a transformative institution in Indian mathematics, striving to recapture historical prestige and propel Indian mathematical research onto the global stage through focused, collaborative scholarship.[1]

Further reading

- [1] Akanksha Mishra, “India’s got a glorious math history, now it will have a future. Lodha s getting the best brains” from *The Print*, 25 August 2025,

<https://theprint.in/ground-reports/indias-got-a-glorious-math-history-now-it-will-hav>

(This is a summary of the original article, produced using GPT-4o.)

OpenAI bets big on India

Adapted from: “OpenAI bets big on India with Rs. 4.5 cr IIT-Madras grant, 5L free ChatGPT licenses.” *Times of India*, 25 August 2025.
https://timesofindia.indiatimes.com/technology/tech-news/openai-bets-big-on-india-with-rs-4-5cr-iit-madras-grant-5l-free-chatgpt-licenses/articleshow/123503858.cms?utm_source=chatgpt.com. Author: Manash Pratim Gohain

OpenAI has launched an “India-first” Learning Accelerator, marked by a Rs. 4.5 cr research grant to IIT-Madras and the distribution of 500,000 free ChatGPT licenses for teachers and students in government schools and AICTE-regulated institutions. Recognizing India as a “lighthouse” for its global education strategy, OpenAI aims to expand access, training, and research in AI-enabled education through partnerships with leading institutions.[1]

Key Objectives and Initiatives

A central component of the programme is a research partnership with IIT-Madras, supported by the grant, to examine how AI affects teaching methods, learning outcomes, and cognitive development. Free six-month ChatGPT licenses will give teachers premium access for lesson planning, assignments, and student engagement, while students will benefit from advanced features like Study Mode and interactive quizzes. These initiatives are intended to narrow India's digital divide, leveraging ChatGPT's compatibility with basic smartphones and support for 11 Indian languages and voice interaction—targeting government schools and cost-effective technical institutions.

Inclusive Approach and Research Collaboration

India already boasts the world's largest ChatGPT student population, using the tool for homework, exam preparation, and explorations of new ideas. OpenAI emphasizes using AI to enrich learning instead of encouraging shortcuts, aiming to empower educators and learners by working collaboratively with institutions. IIT-Madras will lead research, sharing findings publicly to inform teaching practices and future AI product development. Free ChatGPT licenses will be distributed via the Ministry of Education, AICTE, and ARISE, with training programmes developed to build AI literacy and confidence among educators and students.

National Impact and Expansion

OpenAI's expanded presence in India includes a new Delhi office, the launch of ChatGPT Go with UPI payments, OpenAI Academy literacy initiatives with MeitY, and enhanced Indic-language features in GPT-5. By positioning India as a testbed for global implementation, OpenAI hopes this initiative will guide worldwide efforts to use AI effectively in education.[1]

Further reading

- [1] Manash Pratim Gohain, "OpenAI bets big on India with Rs. 4.5 cr IIT-Madras grant, 5L free ChatGPT licenses" from *Times of India*.
<https://timesofindia.indiatimes.com/technology/tech-news/openai-bets-big-on-india-with-rs-4-5cr-iit-madras-grant-5l-free-chatgpt-licenses/articleshow/12350.cms>

(This is a summary of the original article, produced using GPT-4o.)

China is driving renewable energy growth globally

Adapted from: “China is driving renewable energy growth globally – it’s also the world’s biggest coal producer.” <https://sherwood.news/world/china-is-driving-renewable-energy-growth-globally-its-also-the-worlds/>.

Author: Millie Giles

In 2025, global renewable energy investment is at a record high, reaching \$386 billion in the first half of the year, but the US has significantly reduced its spending on wind and solar energy. US investment in renewables dropped by \$20.5 billion (36%) compared to late 2024, with wind investments alone falling 67% year-over-year. This decline follows the Trump administration’s rollback of subsidies for renewable energy, including cuts in July’s “One Big Beautiful Bill Act,” which slashed funding for solar and wind projects.



Blue solar photovoltaic panels neatly arranged in Yuncheng City (CFOTO/Getty Images)

Figure 29

While US investments falter, China is driving global renewable energy growth. Despite a slight contraction in its spending, China still accounted for 44% of global renewable energy investment in H1 2025. It continues to expand its capacity, recently unveiling plans for the world's largest solar farm, covering 235 square miles. As the world's largest energy consumer, China is positioned at both extremes of the energy spectrum: it is the biggest producer of clean energy and the largest carbon emitter. Although China has pledged to peak carbon emissions before 2030 and achieve carbon neutrality by 2060, it is also rapidly expanding coal power, with 94 gigawatts of new coal capacity under construction in 2024.

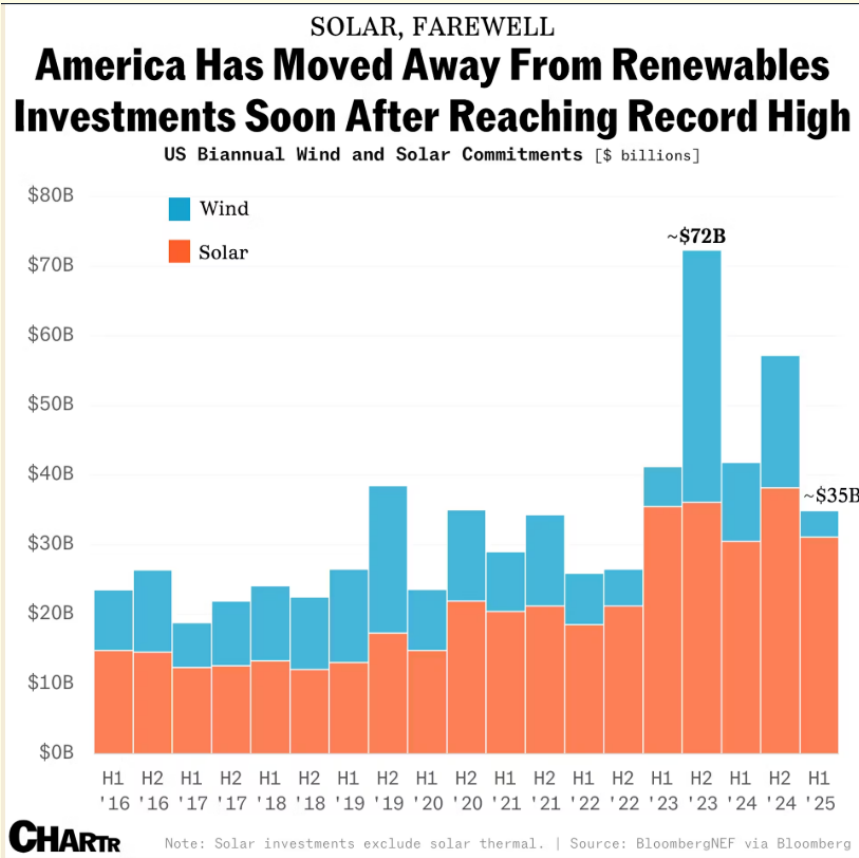


Figure 30

This dual approach highlights the contradictions in China’s energy strategy, balancing its ambitious renewable energy goals with an ongoing reliance on coal and gas, driven partly by rising energy demands from AI and other

industries. The global energy landscape remains polarized, with China leading renewable advancements while the US steps back from previous commitments. How these trends unfold will shape the future of energy and climate action worldwide.

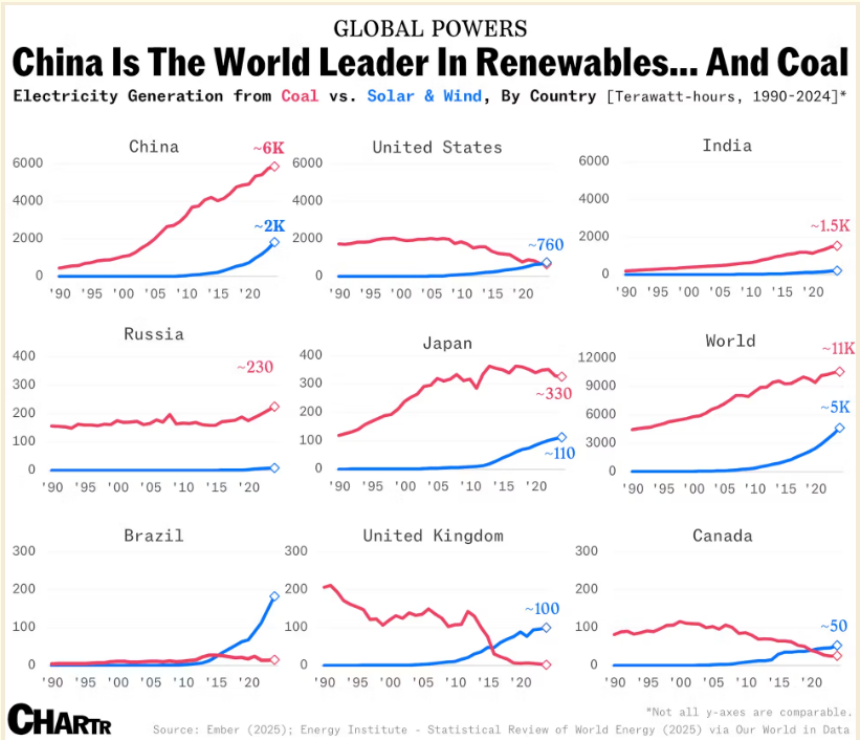


Figure 31

(This is a summary of the original article, produced using GPT-4o.)

The Luhn algorithm

Adapted from: (a) https://en.wikipedia.org/wiki/Luhn_algorithm and (b) <https://www.scientificamerican.com/article/what-is-the-uhn-algorithm-the-math-behind-secure-credit-card-numbers/>. Authors: (a) Contributors to Wikimedia projects (b) Jack Murtagh edited by Jeanna Bryner

The Luhn algorithm is a method used to detect typographical errors in credit card numbers and other such identifiers. It was developed by IBM researcher Hans Peter Luhn in 1960. Its purpose is to catch the most common errors when entering card numbers, without needing to check against a master list or contact the bank instantly.

Here are some remarks from the two articles.

- If you mess up any single digit when entering your card number, the Luhn algorithm will detect it. If you accidentally swap adjacent card digits while inputting, it will detect that, too (with the one exception of flipping 09 to 90, or vice versa).

- Passing Luhn's algorithm does not guarantee a valid credit card number. Rather, failing it guarantees an invalid credit card number. The algorithm puts up a first line of defence that less common typos and clever fraudsters can slip through.

To understand how the algorithm works, you must first know the digital structure of a credit card number.

Structure of a Credit Card Number

Credit card numbers generally comprise the following:

- The major industry identifier (first digit, e.g., Visa = 4)
- The bank identifier (next five to seven digits)
- Individual account number (all the remaining digits except the last)
- Check digit (the final digit, determined by the Luhn algorithm)

How the Luhn Algorithm Works

- Write out all but the last digit of the card number.
- Double every second number starting at the right.
- Sum the digits of the resulting numbers (sum the *digits*, not the numbers; so, if a number doubles to two digits, add them separately; e.g., 7 becomes 14 on doubling, yielding $1 + 4 = 5$).
- Add the check digit to the sum. If the result is a multiple of 10, declare the number to be valid. Else declare the number to be invalid.

Algorithm Features & Error Detection

- The algorithm catches any single-digit mistake and most adjacent digit swaps (except for 09/90).
- These two types of errors have been found to account for nearly 90% of human input mistakes.

Example computation

The article (b) next walks through checking a sample Visa card number, showing how to double and sum the digits, then add the check digit — demonstrating error detection.

Let's suppose that the following number has been entered by the customer when asked to give his or her credit card number:

4024 6072 3695 0748.

Here the last digit (8) is the 'check digit'.

Let's apply the algorithm to this number.

- Double every digit starting from the right (leaving out the check digit 8). We get:

8 0 **4** 4 **12** 0 **14** 2 **6** 6 **18** 5 **0** 7 **8**.

We have highlighted the results of the doubling. (The non-highlighted digits are the same as in the original number.)

- Now add all the digits in the resulting string. We get:

$$8 + 0 + 4 + 4 + 1 + 2 + 0 + 1 + 4 + 2 + 6 + 6 + 1 + 8 + 5 + 0 + 7 + 8 = 67.$$

- Now add the check digit to this:

$$67 + 8 = 75.$$

- Is the final answer a multiple of 10? No, it isn't. *So the number entered by the customer is not valid.*

Efficiency & Use

The algorithm is simple and fast; it lets businesses and websites detect common errors without costly validation delays.

Because the algorithm requires such little processing power, the computer handling the transaction can also run the check without needing to contact any third party.

The article explains with examples why swapping digits alters the sum, making it highly sensitive to simple mistakes. This is an important feature of the algorithm.

Limitations

The algorithm is not foolproof against clever fraud or rarer typo patterns; more rigorous bank validation is still needed.

Closing remark

The Luhn algorithm is an elegant mathematical safeguard that efficiently catches the vast majority of real-world input errors in credit card numbers and other similar identifiers, e.g., barcodes.

Pseudocode for the algorithm

The Wikipedia article (a) gives pseudocode for implementation of the Luhn algorithm:

```
function isValid(cardNumber[1..length])
    sum := 0
    parity := length mod 2
    for i from 1 to (length - 1) do
        if i mod 2 == parity then
            sum := sum + cardNumber[i]
        elseif cardNumber[i] > 4 then
            sum := sum + 2 * cardNumber[i] - 9
        else
            sum := sum + 2 * cardNumber[i]
        end if
    end for
    return cardNumber[length] == ((10 - (sum mod 10)) mod 10)
end function
```

(This is a summary of the original two articles, produced using the LLM Perplexity.)

A ‘last digit’ problem

In this short article we pose and solve the following problem:

Problem Show that the number

$$\left(\sqrt{2} + 1\right)^{2025} - \left(\sqrt{2} - 1\right)^{2025}$$

is an integer, and find its units digit.

Solution

To show that the given quantity is an integer it’s fairly easy. We only need to use the binomial theorem to expand the two expressions. We ask you to do this verification by yourself.

Let us now focus our attention on the question about the units digit. The exponent 2025 is surely a ‘red herring’. No one would want compute a quantity that has to be raised to that power! Therefore, the question more or less invites us to examine more generally the following sequence:

$$a_n = \left(\sqrt{2} + 1\right)^n - \left(\sqrt{2} - 1\right)^n. \quad (40)$$

Let's compute the first few values by hand. We immediately notice an interesting pattern. It appears that a_n is an integer for all *odd* values of n , and an integer multiple of $\sqrt{2}$ for all even values of n :

n	1	3	5	7	9	11	13	15	
a_n	2	14	82	478	2786	16238	94642	551614	...

and

n	2	4	6	8	10	12	
a_n	$4\sqrt{2}$	$24\sqrt{2}$	$140\sqrt{2}$	$816\sqrt{2}$	$4756\sqrt{2}$	$27720\sqrt{2}$	...

Do you notice a striking recurrence in both the sequences? In each, it appears that a_n is equal to 6 times the previous value minus the value previous to that (e.g., $82 = 6 \times 14 - 2$). In other words, it seems to be the case that

$$a_n = 6a_{n-2} - a_{n-4}, \quad \text{for all } n > 4. \quad (41)$$

This seems to be true for both odd n and even n . Can we be sure that it is so? Yes — but (naturally) we have to prove it first! There is an easy proof by induction. We invite you to complete the proof on your own.

Reformulation of the problem

A slight reformulation of the problem makes it easier to analyze, and that's what we'll do.

We consider this related sequence,

$$c_n = \left(1 + \sqrt{2}\right)^n + \left(1 - \sqrt{2}\right)^n. \quad (42)$$

It is easy to see that $c_n = a_n$ for odd values of n . Calculations suggest that c_n is an integer for *all* positive integers n :

n	1	2	3	4	5	6	7	8	9	10	...
c_n	2	6	14	34	82	198	478	1154	2786	6726	...

Similar to the recurrence relation for a_n , we spot a very simple recurrence relation for c_n :

$$c_n = 2c_{n-1} + c_{n-2}, \quad \text{for all } n > 2. \quad (43)$$

We choose to prove (43) rather than (41). Our method of proof will be to use the theory of quadratic equations. If you have not seen this kind of an approach earlier, it will be well worth a close study.

Let $u = 1 + \sqrt{2}$ and $v = 1 - \sqrt{2}$. Then

$$u + v = 2, \quad uv = -1,$$

so u and v are the roots of the quadratic equation $x^2 - 2x - 1 = 0$, which we choose to write as

$$x^2 = 2x + 1. \quad (44)$$

Since u and v are the roots of (44), we can say that

$$u^2 = 2u + 1,$$

$$v^2 = 2v + 1.$$

Multiplying through by u^{n-2} and v^{n-2} , respectively we deduce that for any integer $n > 2$,

$$u^n = 2u^{n-1} + u^{n-2},$$

$$v^n = 2v^{n-1} + v^{n-2}.$$

Hence by addition we get

$$c_n = 2c_{n-1} + c_{n-2}, \quad \text{for all } n > 2, \quad (45)$$

just as we wanted.

Since we are interested only in the units digit of c_n , let us read the above relation mod 10:

$$c_n \equiv 2c_{n-1} + c_{n-2} \pmod{10}, \quad \text{for all } n > 2. \quad (46)$$

Using either (46) or (45), we compute a table of values of the units digit of c_n . Let d_n denote the units digit of c_n . Here's what we obtain:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...
d_n	2	6	4	4	2	8	8	4	6	6	8	2	2	6	4	...

We see a clear pattern. The units digit appears to cycle through a string with length 12, containing these digits:

$$2, 6, 4, 4, 2, 8, 8, 4, 6, 6, 8, 2, \dots \quad (47)$$

Following this we get another repetition of 2,6,4,4,2,8,8,4,6,6,8,2 (the same string), and so on, indefinitely.

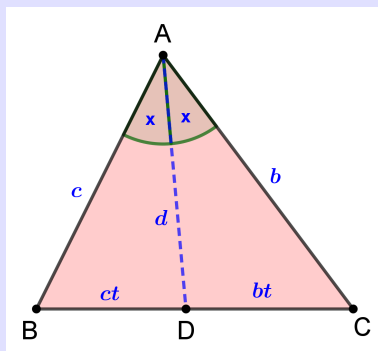
Relation (46) implies that at any point, if we get two successive values of d_n that have occurred earlier (as two successive values), then the sequence will repeat from that point on. And this is just what we observe. After the initial 2 and 6 at positions 1 and 2, we observe a 2 and 6 again at positions 13 and 14. So the string will repeat with a cycle length of 12.

The problem asks us to compute d_{2025} . Since $2025 \equiv 9 \pmod{12}$, it follows that $d_{2025} = d_9$. Hence $d_{2025} = 6$. Thus the required units digit is 6. ■

The Angle Bisector Theorem – Part 3

This is a continuation of the articles in the last two issue dealing with the angle bisector theorem.

In this article, we consider a related problem: finding the *length of the angle bisector* in terms the sides of the triangle.



In $\triangle ABC$, let AD be the bisector of $\angle BAC$. We wish to find the length d of AD in terms of a, b, c .

We find the lengths of BD and CD by using the angle bisector theorem, which tells us that $BD/DC = c/b$, so $BD = ct$ and $CD = bt$ for some constant t .

Figure 32

Since $BD + DC = a$, we get

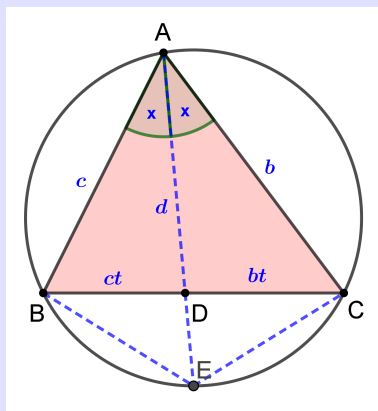
$$t(b+c) = a, \quad \therefore t = \frac{a}{b+c},$$

hence,

$$BD = \frac{ac}{b+c}, \quad CD = \frac{ab}{b+c}. \quad (48)$$

This is an important and much-used result.

An additional construction is now needed.



Extend the angle bisector AD till it meets the circumcircle of ABC at E (say). We find the length of AD by using the chord theorem (discussed elsewhere in this issue). We have,

$$AD \times DE = BD \times CD$$

$$= \frac{a^2 bc}{(b+c)^2}. \quad (49)$$

Figure 33

Next, $\triangle ABD \sim \triangle AEC$ (check the equality of the angles of the triangle to see why). Hence,

$$\frac{AD}{AC} = \frac{AB}{AE},$$

$$\therefore AD \times AE = AB \times AC = bc. \quad (50)$$

Subtraction, (50) minus (49), yields the desired formula,

$$AD^2 = bc \left(1 - \frac{a^2}{(b+c)^2} \right). \quad (51)$$

An expression for d using trigonometric ratios may also be found by using the sine rule. In $\triangle ABD$ we have

$$\frac{AD}{\sin B} = \frac{BD}{\sin A/2}, \quad \therefore d = \frac{ac}{b+c} \cdot \frac{\sin B}{\sin A/2}.$$

Since $a \sin B = b \sin A$, we may rewrite this as

$$d = \frac{bc}{b+c} \cdot \frac{\sin A}{\sin A/2}.$$

Hence,

$$d = \frac{2bc}{b+c} \cdot \cos \frac{A}{2}. \quad (52)$$

The Steiner-Lehmus problem

The Steiner-Lehmus problem is a notorious problem of Euclidean geometry. Here is its statement:

Problem. Show that in a triangle if two of the angle bisectors have equal length, then the triangle is isosceles.

The problem gives the appearance of having a simple solution, merely requiring some pair of triangles to be proved congruent to each other. After initial attempts we discover that the configuration does not quite fit the requirements of any of the congruence theorems (SAS congruence, ASA congruence, etc). Pure geometry solutions to the problem *have* been found — but they are not easy to find! For now, we will only present an algebraic solution based on the formula for the length of the angle bisector (found above). We will study some pure geometry solutions in later issues.

Algebraic solution to the Steiner-Lehmus problem

We shall make use of formula (51):

$$d^2 = bc \left(1 - \frac{a^2}{(b+c)^2} \right).$$

Suppose that in $\triangle ABC$, the length of the angle bisector of $\angle B$ is equal to the length of the angle bisector of $\angle C$. Then we have,

$$ac \left(1 - \frac{b^2}{(a+c)^2} \right) = ab \left(1 - \frac{c^2}{(a+b)^2} \right). \quad (53)$$

Let us now bring all the terms of (53) to one side, clear the fractions by multiplying through by $(a+b)^2(a+c)^2$, and then factorize the resulting expression to the extent possible. We get,

$$ac \left((a+c)^2 - b^2 \right) (a+b)^2 - ab \left((a+b)^2 - c^2 \right) (a+c)^2 = 0,$$

and therefore,

$$a(a+b+c)(c-b) \left(a^3 + a^2b + a^2c + b^2c + bc^2 + 3abc \right) = 0. \quad (54)$$

The first factor, a , is readily visible, while the second factor, $a+b+c$, comes from the “difference of two squares” formula. These are easy to figure out, but you may wonder how we found the factor $c-b$.

Look at it this way: $c-b$ *must* be a factor, because equality (53) is clearly true if $b=c$. Now we only need to invoke the factor theorem. So actually, there is no great mystery involved.

What will make equality (54) true? Let’s explore ...

The left side of (54) is a product of four factors and will be zero only if one of the four factors is 0. So, the possibilities are:

(i) $a = 0$

(ii) $a + b + c = 0$

(iii) $c - b = 0$

(iv) $a^3 + a^2b + a^2c + b^2c + bc^2 + 3abc = 0$.

Possibilities (i) and (ii) can be ruled out because $a > 0$, $a + b + c > 0$, and (iv) can be ruled out as well because $a^3 + a^2b + a^2c + b^2c + bc^2 + 3abc$ is a sum of positive quantities (since $a, b, c > 0$) and must itself be positive.

The only possibility left is (iii). Hence, $b = c$.

We conclude that if the lengths of the bisectors of $\angle B$ and $\angle C$ are equal, then $b = c$. This proves the Steiner-Lehmus theorem. ■

Trigonometric solution to the Steiner-Lehmus problem

As an alternative to the solution using algebra (a lot of algebra!), here is a solution that uses trigonometric ratios. Some readers may prefer this proof!

From (52), we know that the lengths of the bisectors of $\angle B$ and $\angle C$ are

$$\frac{2ac}{a+c} \cdot \cos \frac{B}{2}, \quad \frac{2ab}{a+b} \cdot \cos \frac{C}{2}. \quad (55)$$

We must explore what makes these two quantities equal. Removing the common factor $2a$, we see that we must compare the following two quantities:

$$\frac{c}{a+c} \cdot \cos \frac{B}{2}, \quad \frac{b}{a+b} \cdot \cos \frac{C}{2}. \quad (56)$$

Now suppose that $b > c$.

Then $\angle B > \angle C$, hence $B/2 > C/2$, hence $\cos B/2 < \cos C/2$. We can be sure that this implication is true, because angles $B/2$ and $C/2$ are both acute. (Remember, the cosine function is strictly decreasing between 0° and 90° .)

Also, if $b > c$, then

$$\frac{c}{a+c} < \frac{b}{a+b}.$$

Here is a short proof of this claim:

$$\begin{aligned} \frac{b}{a+b} - \frac{c}{a+c} &= \frac{b(a+c) - c(a+b)}{(a+b)(a+c)} \\ &= \frac{a(b-c)}{(a+b)(a+c)} > 0. \end{aligned}$$

So, if $b > c$, then we have

$$\frac{c}{a+c} < \frac{b}{a+b} \quad \text{and} \quad \cos \frac{B}{2} < \cos \frac{C}{2}. \quad (57)$$

So, if $b > c$, then the two quantities in (55) cannot be equal. Symmetrically, the same is true if $c > b$.

It follows that if those two quantities are equal, then we must have $b = c$. This means that equality of the lengths of the two angle bisectors implies that the triangle is isosceles. ■

We will study some pure geometry proofs of this Steiner-Lehmus theorem in the next issue.

How many triangles are there with a given perimeter?

Article contributed by Shailesh Shirali

In this article we study the following combinatorial problem: *How many integer-sided triangles are there whose perimeter is a given positive integer n ?*

For example, if we specify that the perimeter is 5, we find after some reflection that there is just one integer-sided triangle that fits this constraint: the triangle with sides 2,2,1. Check this claim out for yourself! You'll find that it is so.

Let $f(n)$ denote the required number. Trivially we have $f(1) = 0 = f(2)$, as there cannot be any integer-sided triangles with perimeter 1 or 2; and $f(3) = 1$ (there is just one integer-sided triangle with perimeter 3, with sides 1,1,1).

What is the value of $f(4)$? Let the sides be a, b, c , with $a \geq b \geq c > 0$. We must be careful not to count any triangle more than once; that's why we

fix these inequalities. So we have,

$$a + b + c = 4, \quad a \geq b \geq c > 0.$$

There is just one solution to this system: $(a, b, c) = (2, 1, 1)$. But there is no triangle whose sides have lengths 2, 1, 1, as these numbers do not satisfy the triangle inequality! Remember that in a triangle, the sum of any two sides must exceed the third side. This means that there are no integer-sided triangles with perimeter 4. What a surprise! — it means that $f(4) = 0$.

Let's set up the general problem algebraically. Let the sides be a, b, c with $a \geq b \geq c$. Of course, we must have $c > 0$. The triangle inequality tells us that we must also have

$$a + b > c, \quad a + c > b, \quad b + c > a.$$

Since we have assumed that $a \geq b \geq c > 0$, the first two inequalities are automatically satisfied. So we need to only ensure that the third condition holds.

It follows that $f(n)$ is equal to the number of integer triples (a, b, c) satisfying the following conditions:

$$a \geq b \geq c > 0, \quad a + b + c = n, \quad b + c > a.$$

We can reduce the dimensionality of the problem from 3 to 2 by making use of condition $a + b + c = n$, which yields $c = n - a - b$. This leads to the following system:

$$a \geq b \geq n - a - b > 0,$$

$$b + (n - a - b) > a,$$

i.e.,

$$a \geq b,$$

$$a + 2b \geq n,$$

$$a + b < n,$$

$$2a < n.$$

Now we can eliminate the third condition, for if $a < n/2$ and $b \leq a$, then surely $a + b < n$. So we arrive at a compact description: $f(n)$ is equal to the number of positive integer pairs (a, b) such that

$$b \leq a < \frac{n}{2}, \quad a + 2b \geq n.$$

From $a + 2b \geq n$ we get $a + 2a \geq n$ (since $a \geq b$), i.e., $3a \geq n$. Hence,

$$a \geq \frac{n}{3}.$$

Let's see if this formulation helps us to compute $f(n)$ for some sample values of n . Let's compute $f(10)$. We have, since the integer just greater than $10/3$ is 4,

$$a \geq 4, \quad b \leq a < 5, \quad a + 2b \geq 10.$$

From the first two constraints we see that $a = 4$ (no other value is possible). The constraints on b are now,

$$b \leq 4, \quad 2b \geq 6, \text{ i.e., } b \geq 3.$$

There are just two possible values that b can take: $b = 3, b = 4$. Hence the possible (a, b) pairs are $(4, 3)$ and $(4, 4)$. The first one yields $c = 10 - 7 = 3$, while the second one yields $c = 10 - 8 = 2$. We see that there are just

two integer-sided triangles with perimeter 10, with sides 4,4,2 and 4,3,3 respectively. Hence $f(10) = 2$.

Let's repeat the computation for a larger value of n , say $n = 40$. The conditions are now,

$$a \geq \frac{40}{3}, \quad b \leq a < 20, \quad a + 2b \geq 40,$$

i.e.,

$$a \geq 14, \quad b \leq a \leq 19, \quad b \geq 20 - \frac{a}{2}.$$

The possible values for a are 14, 15, ..., 18, 19. Let's work out the range of values for b for each value of a .

- $a = 19$ leads to $b \leq 19$ and $b \geq 11$, hence 9 possibilities
- $a = 18$ leads to $b \leq 18$ and $b \geq 11$, hence 8 possibilities
- $a = 17$ leads to $b \leq 17$ and $b \geq 12$, hence 6 possibilities
- $a = 16$ leads to $b \leq 16$ and $b \geq 12$, hence 5 possibilities
- $a = 15$ leads to $b \leq 15$ and $b \geq 13$, hence 3 possibilities
- $a = 14$ leads to $b \leq 14$ and $b \geq 13$, hence 2 possibilities.

Hence $f(40) = 9 + 8 + 6 + 5 + 3 + 2 = 33$.

We may express the formula for $f(n)$ even more compactly as a summation:

$$f(n) = \sum_{a=\lceil n/3 \rceil}^{a=\lfloor (n-1)/2 \rfloor} \left(a + 1 - \left\lceil \frac{n-a}{2} \right\rceil \right).$$

This may look forbidding — but it amounts to exactly the same thing as the previous formulation. See if you can decipher the formula!

Let's use the summation formula to compute $f(100)$. Since $\lceil 100/3 \rceil = 34$ and $\lfloor 99/2 \rfloor = 49$, we have,

$$\begin{aligned} f(100) &= \sum_{a=34}^{a=49} \left(a + 1 - \left\lceil \frac{100 - a}{2} \right\rceil \right) \\ &= \sum_{a=34}^{a=49} (a + 1) - \sum_{a=34}^{a=49} \left\lceil \frac{100 - a}{2} \right\rceil. \end{aligned}$$

The first part of the summation in the last line is

$$\sum_{a=34}^{a=49} (a + 1) = 35 + 36 + \cdots + 49 + 50 = \frac{35 + 50}{2} \times 16 = 680.$$

The second part of the summation is

$$\begin{aligned} \sum_{a=34}^{a=49} \left\lceil 50 - \frac{a}{2} \right\rceil &= 33 + 33 + 32 + 32 + 31 + 31 + \cdots + 26 + 26 \\ &= 2(26 + 27 + \cdots + 32 + 33) \\ &= (26 + 33) \times 8 = 472. \end{aligned}$$

Hence $f(100) = 680 - 472 = 208$.

The last formulation makes the computation quite easy. You may wonder whether there is any 'nice' and simple formula for $f(n)$. The answer is: not quite, but we come close.

Armed with the formulation displayed above, we prepare a table of values of $f(n)$, in an effort to study its patterns. For reasons that we shall explain shortly, we opt to display the successive values in sets of 12. We obtain the following table.

1 to 12	0	0	1	0	1	1	2	1	3	2	4	3
13 to 24	5	4	7	5	8	7	10	8	12	10	14	12
25 to 36	16	14	19	16	21	19	24	21	27	24	30	27
37 to 48	33	30	37	33	40	37	44	40	48	44	52	48
49 to 60	56	52	61	56	65	61	70	65	75	70	80	75
61 to 72	85	80	91	85	96	91	102	96	108	102	114	108
73 to 84	120	114	127	120	133	127	140	133	147	140	154	147
85 to 96	161	154	169	161	176	169	184	176	192	184	200	192
97 to 108	208	200	217	208	225	217	234	225	243	234	252	243

We now explain why we have displayed the successive f -values in sets of 12. Look again at formulation where we expressed $f(n)$ as a summation:

$$f(n) = \sum_{a=\lceil n/3 \rceil}^{a=\lfloor (n-1)/2 \rfloor} \left(a + 1 - \left\lceil \frac{n-a}{2} \right\rceil \right).$$

We see that the summation limits involve denominators of 3 (lower limit) and 2 (upper limit). Now, division by 3 yields remainders that cycle through 3 values, and division by 2 yields remainders that cycle through 2 values. This suggests that the sequence of f -values will be periodic with cycle length 6. Inside the summation sign we see a denominator of 2. This could integrate with previously mentioned periodicity, thereby maintaining a cycle length of 6. Or it may give rise to periodicity with cycle length 12. This line of reasoning suggests that if we list the f -values in sets of 12, we may hit upon the underlying periodicity more readily. Hence our choice of display.

And this choice yields dividends immediately! We find that each column of values, considered as a sequence by itself, has a visible periodicity. For example, consider the numbers in the first column; these are the values of $f(1), f(13), f(25), f(37), f(49), \dots$:

$$0, \quad 5, \quad 16, \quad 33, \quad 56, \quad 85, \quad 120, \quad 161, \quad 208, \quad \dots$$

Here are the first differences of this sequence:

$$5, \quad 11, \quad 17, \quad 23, \quad 29, \quad 35, \quad 41, \quad 47, \quad \dots$$

And here are the second differences:

$$6, \quad 6, \quad 6, \quad 6, \quad 6, \quad 6, \quad \dots$$

— a series of constants! The fact that the second differences are a constant tells us that we're dealing with a *quadratic sequence* (degree 2). We can make this more precise. The numbers $1, 13, 25, 37, 49, \dots$ are of the form $12k + 1$. Let $a_k = f(12k + 1)$. Then we have,

$$a_1 - a_0 = 5,$$

$$a_2 - a_1 = 11,$$

$$a_3 - a_2 = 17,$$

$$a_4 - a_3 = 23, \quad \dots,$$

and in general,

$$a_k - a_{k-1} = 5 + 6(k - 1),$$

i.e., $a_k - a_{k-1} = 6k - 1$.

Hence,

$$\begin{aligned} a_k &= 6k - 1 + 6(k - 1) - 1 + 6(k - 2) - 1 + \cdots + 6(1) - 1 \\ &= 6 \times \frac{k(k+1)}{2} - k = 3k(k+1) - k, \end{aligned}$$

so $a_k = 3k^2 + 2k$. Therefore we have,

$$f(12k + 1) = 3k^2 + 2k.$$

A whole cluster of such results may be obtained; for example,

$$f(12k + 2) = 3k^2 + k,$$

$$f(12k + 3) = 3k^2 + 3k + 1,$$

$$f(12k + 4) = 3k^2 + 2k,$$

and so on.

We invite you to work out the corresponding formula for each column.

You will find that the term common to all the formulas is $3k^2$. Since

$$3k^2 = \frac{(12k)^2}{48},$$

it means that to a first approximation,

$$f(n) \approx \frac{n^2}{48},$$

and so,

$$\lim_{n \rightarrow \infty} \frac{f(n)}{n^2/48} = 1.$$

We invite the reader to explore the problem further.

Inverting the Perspective

Article contributed by Sudarshan Kasturi

In this article, I demonstrate a beautiful and useful idea in problem solving. We refer to it as “inverting the perspective.” The idea is to see a problem in more than one way. This can lead to unexpectedly beautiful results. I consider examples from different fields of mathematics.

An example from Geometry

Normally, we use the Pythagorean theorem to prove the Cosine Rule which is then used to obtain Heron’s formula for the area of a triangle. Let us reverse the situation and see if applying Heron’s formula in a right-angled triangle gives us the famous theorem due to Pythagoras.

Consider a right triangle, right-angled at C , with AB as the hypotenuse. Its area is then given by

$$[\triangle ABC] = \frac{1}{2}ab. \quad (58)$$

Heron's formula tells us that the area is also given by

$$[\triangle ABC] = \sqrt{s(s-a)(s-b)(s-c)}, \quad (59)$$

where $2s = a + b + c$ is the perimeter of the triangle.

Thus, we are looking at the same entity, namely $[\triangle ABC]$, in two different ways. It follows that

$$\sqrt{s(s-a)(s-b)(s-c)} = \frac{1}{2}ab,$$

$$\therefore 2s(2s-2a)(2s-2b)(2s-2c) = 4a^2b^2,$$

$$\therefore (a+b+c)(b+c-a)(a+c-b)(a+b-c) = 4a^2b^2,$$

$$\therefore \left((a+b)^2 - c^2\right) \left(c^2 - (a-b)^2\right) = 4a^2b^2,$$

$$\therefore c^4 - 2c^2(a^2 + b^2) + (a^2 - b^2)^2 = -4a^2b^2,$$

$$\therefore c^4 - 2c^2(a^2 + b^2) + (a^2 - b^2)^2 + 4a^2b^2 = 0,$$

$$\therefore c^4 - 2c^2(a^2 + b^2) + (a^2 + b^2)^2 = 0,$$

$$\therefore \left(c^2 - (a^2 + b^2)\right)^2 = 0,$$

$$\therefore a^2 + b^2 = c^2,$$

which is the statement of the Pythagorean theorem. ■

Notice how the main idea used was looking at the same thing (area) in two different ways.

An example from Algebra

We begin by establishing an algebraic identity.

Consider the sum

$$S = 1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1}, \quad (60)$$

where n is any positive integer and $0 < x < 1$. Multiplying by x , we get,

$$S = 1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1}, \quad (61)$$

$$xS = x + 2x^2 + 3x^3 + 4x^4 + \cdots + nx^n. \quad (62)$$

Subtraction yields,

$$\begin{aligned} (1-x)S &= 1 + x + x^2 + \cdots + x^{n-1} - nx^n = \frac{1-x^n}{1-x} - nx^n \\ &= \frac{1 - (n+1)x^n + nx^{n+1}}{(1-x)}, \\ \therefore S &= \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}. \end{aligned} \quad (63)$$

Now put $y = x - 1$ in (60) and (63). We get,

$$S = 1 + 2(1+y) + 3(1+y)^2 + 4(1+y)^3 + \cdots + n(1+y)^{n-1}, \quad (64)$$

$$S = \frac{n(1+y)^{n+1} - (n+1)(1+y)^n + 1}{y^2}. \quad (65)$$

From (64) and (65), we get,

$$\begin{aligned} &1 + 2(1+y) + 3(1+y)^2 + 4(1+y)^3 + \cdots + n(1+y)^{n-1} \\ &= \frac{n(1+y)^{n+1} - (n+1)(1+y)^n + 1}{y^2}. \end{aligned} \quad (66)$$

We now obtain the coefficient of the constant term on both sides of (66). That is, we try to look at the same thing in two different ways.

The constant term on the left side of (66) is $1 + 2 + 3 + \cdots + n$.

The constant term on the right side of (66) is the coefficient of y^2 in the numerator of the fraction. This is equal to

$$\begin{aligned} & n \binom{n+1}{2} - (n+1) \binom{n}{2} \\ &= n \frac{(n+1)n}{2} - (n+1) \frac{n(n-1)}{2} \\ &= \frac{(n+1)n}{2}, \quad \text{after simplification.} \end{aligned}$$

Hence, we obtain

$$1 + 2 + 3 + \cdots + n = \frac{(n+1)n}{2}. \quad (67)$$

Next let us look at the coefficient of y on both sides of (66).

On the left side the coefficient is

$$\begin{aligned} & (2 \times 1) + (3 \times 2) + (4 \times 3) + \cdots + n(n-1) \\ &= (1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2) - (1 + 2 + 3 + 4 + \cdots + n) \\ &= (1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2) - \frac{n(n+1)}{2}. \end{aligned} \quad (68)$$

On the right side the coefficient is

$$n \binom{n+1}{3} - (n+1) \binom{n}{3} = \frac{n(n+1)(n-1)}{3}, \quad (69)$$

on simplification. Hence we have

$$\begin{aligned}
 1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 &= \frac{n(n+1)}{2} + \frac{n(n+1)(n-1)}{3} \\
 &= \frac{n(n+1)}{6} (3 + 2(n-1)) \\
 &= \frac{n(n+1)(2n+1)}{6}, \tag{70}
 \end{aligned}$$

a familiar result which we have obtained in a very different way.

Readers are invited to compare the coefficients of y^2 on the two sides of (66). They will arrive at a neat expression for the sum of the cubes of the first n positive integers.

An example from Trigonometry

Consider a unit circle centred at the origin O . Fix a pair of points $A(1,0)$ and $C(-1,0)$, and let B be an arbitrary point on this circle in the first quadrant. Let $\angle BCA = \theta$, so $0 \leq \theta \leq \pi/4$.

Using the cosine rule in $\triangle OAB$, we get

$$AB^2 = 2(1 - \cos 2\theta).$$

At the same time in the right-angled triangle ABC we have,

$$AB = 2\sin\theta.$$

It follows that

$$\begin{aligned}
 2(1 - \cos 2\theta) &= 4\sin^2\theta, \\
 \therefore \cos 2\theta &= 1 - 2\sin^2\theta = 2\cos^2\theta - 1. \tag{71}
 \end{aligned}$$

We have thus obtained the double-angle identities for the cosine function using the cosine rule.

Can we similarly obtain the double angle identity for the sine function using the sine rule? Yes, we can.

Using the sine rule in triangle AOB , we get,

$$\frac{AB}{\sin 2\theta} = \frac{1}{\sin(90^\circ - \theta)} = \frac{1}{\cos \theta},$$

$$\therefore \sin 2\theta = AB \cdot \cos \theta = 2 \sin \theta \cdot \cos \theta. \quad (72)$$

Questions for the reader: Where have we assumed that θ is acute? Can these results be extended to non-acute angles too?

Challenge for the reader: Obtain expressions for $\sin 3\theta$ and $\cos 3\theta$ in a similar way. (You may wish to consider a triangle with angles θ and 2θ .)

An example from Calculus

Consider the function

$$y = \arcsin \frac{x-1}{x+1}.$$

Differentiating, using the Chain Rule and simplifying gives,

$$\frac{dy}{dx} = \frac{1}{(x+1)\sqrt{x}}.$$

This implies that

$$\int \frac{1}{(x+1)\sqrt{x}} dx = \arcsin \frac{x-1}{x+1} + c, \quad (73)$$

where c is an arbitrary constant.

Let us now try to obtain the integral on the left side in another way.

We see easily by putting $x = t^2$ that $dx = 2t dt$, and so,

$$\begin{aligned}\int \frac{1}{(x+1)\sqrt{x}} dx &= \int \frac{2t}{(t^2+1)t} dt \\ &= 2 \int \frac{1}{t^2+1} dt = 2 \arctan t + d \\ &= 2 \arctan \sqrt{x} + d,\end{aligned}\tag{74}$$

where d is an arbitrary constant. From (73) and (74) we get, by putting $x = 0$,

$$\begin{aligned}\arcsin \frac{x-1}{x+1} + c &= 2 \arctan \sqrt{x} + d, \\ \therefore \arcsin(-1) + c &= 2 \arctan 0 + d, \\ \therefore c - \frac{\pi}{2} &= d.\end{aligned}$$

Hence we obtain,

$$\arcsin \frac{x-1}{x+1} = 2 \arctan \sqrt{x} - \frac{\pi}{2}.\tag{75}$$

Questions for the reader: What are the largest domains of the following functions?

$$f(x) = \arcsin \frac{x-1}{x+1}, \quad g(x) = 2 \arctan \sqrt{x}.$$

Compute the integral

$$\int \frac{1}{(x+1)\sqrt{x}} dx$$

in two different ways (by using two different substitutions) and thereby obtain an identity between $\arcsin(2x - 1)$ and $\arcsin \sqrt{x}$. (Hint: You might have to use the method of “completion of squares” in one of these.)

Closing remark

I hope I have been able to communicate some of the beauty and unexpected applicability of this overarching idea. The reader is invited to explore the questions posed and answer them. I shall be happy to receive any communication in this regard.

Further reading

[1] Dunham, William, *Journey through Genius*, John Wiley & Sons, 1990.

About the author

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Does it ever happen that $2X \neq X + X$?

Article contributed by Shailesh Shirali

Generally, no! In the realm of algebra as we know it, we do not (generally) encounter situations where $2X \neq X + X$.

But there is one situation where it does happen (quite routinely) that $2X \neq X + X$, and that is when X is a random variable.

It is easy to demonstrate this behaviour through an example.

To avoid any misunderstanding, let us first specify what we mean by $2X$ and what we mean by $X + X$ when X is a random variable:

- $2X$ is the random variable whose values are exactly twice those of the random variable X , with the same associated probabilities.
- $X + X$ is the sum of two *independent* random variables each of which

has the same probability distribution as X . The key word here is 'independent'.

Example

Suppose the random variable X has the following probability distribution:

k	0	1
$P(X = k)$	0.5	0.5

The random variable $2X$ will obviously have the following probability distribution:

k	0	2
$P(2X = k)$	0.5	0.5

What about $X + X$? To avoid getting misled by the symbols, it is better to write the expression as $X + Y$, with the understanding that X and Y are *independent*, and Y has the same probability distribution as X .

Since X and Y take values from $\{0, 1\}$ (independently), the range of values of $X + Y$ is $\{0, 1, 2\}$, with 0 arising as $0 + 0$; 1 arising as $1 + 0$ or as $0 + 1$ (so it can arise in two different ways); and 2 arising as $1 + 1$. Hence, the probability distribution of $X + Y$ will be as follows:

k	0	1	2
$P(X + Y = k)$	0.25	0.5	0.25

We see that $2X$ and $X + X$ have different probability distributions. Thus, $2X \neq X + X$ in this case. ■

Remark

For the sake of completeness, let us also compute the expected value and variance of $2X$ and $X + X$. We have,

$$E(2X) = 0(0.5) + 2(0.5) = 1,$$

and

$$E(X + X) = 0(0.25) + 1(0.5) + 2(0.25) = 1.$$

So, $E(2X) = E(X + X)$. We see that the two random variables have the same expected value, though they have different probability distributions.

Does that come as a surprise? Well, it shouldn't! After all, it is known that the expected value of the sum of two random variables is equal to the sum of the individual expected values. (This is a standard result.)

What about the variance? We have,

$$E(2X)^2 = 0(0.5) + 4(0.5) = 2,$$

hence

$$\text{Var}(2X) = 2 - 1^2 = 1,$$

and,

$$E(X + Y)^2 = 0(0.25) + 1(0.5) + 4(0.25) = 2,$$

hence,

$$\text{Var}(X + Y) = 2 - 1^2 = 1.$$

So $2X$ and $X + X$ have equal variance as well. (This too is a standard result. Try to justify it on your own.)

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We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!